

Name: _____

GCSE Biology Unit 1 Foundation Tier

Once you have completed a paper and have marked it, fill out the table below, using the grade boundaries provided to add your grade. Then list which questions you did well and which questions you struggled with. For the questions you struggled with, review why and at the reasons. The most common reasons are

- Not understanding basic concepts (e.g. simple definitions)
- Calculation errors (forgetting to square numbers, etc.)
- Not reading questions carefully
- Not answering the full question (describe vs. explain)
- Laboratory/practical setup mistakes

For these questions it is also useful add notes to the question to help you next time and take a photo of these questions. Use these then to guide further revision.

Year	Score /80	Grade	Good Questions	Bad Questions	Reasons
2017					
2018					
2019					
2022					
2023					

Grade boundaries - Raw Score

Year	G	F	E	D	C	Max UMS
2017	10	14	18	22	27	32
2018	13	18	23	28	34	40
2019	14	18	22	26	30	34
2022	11	16	21	26	31	36
2023	18	22	26	31	36	41



Surname	Centre Number	Candidate Number
Other Names		0



GCSE – **NEW**

3400U10-1



S17-3400U10-1

BIOLOGY – Unit 1:
Cells, Organ Systems and Ecosystems

FOUNDATION TIER

WEDNESDAY, 14 JUNE 2017 – MORNING
1 hour 45 minutes

For Examiner’s use only		
Question	Maximum Mark	Mark Awarded
1.	7	
2.	8	
3.	8	
4.	9	
5.	10	
6.	12	
7.	6	
8.	8	
9.	12	
Total	80	

ADDITIONAL MATERIALS

In addition to this paper you may require a calculator and a ruler.

INSTRUCTIONS TO CANDIDATES

Use black ink or black ball-point pen.
Write your name, centre number and candidate number in the spaces at the top of this page.
Answer **all** questions.
Write your answers in the spaces provided in this booklet.

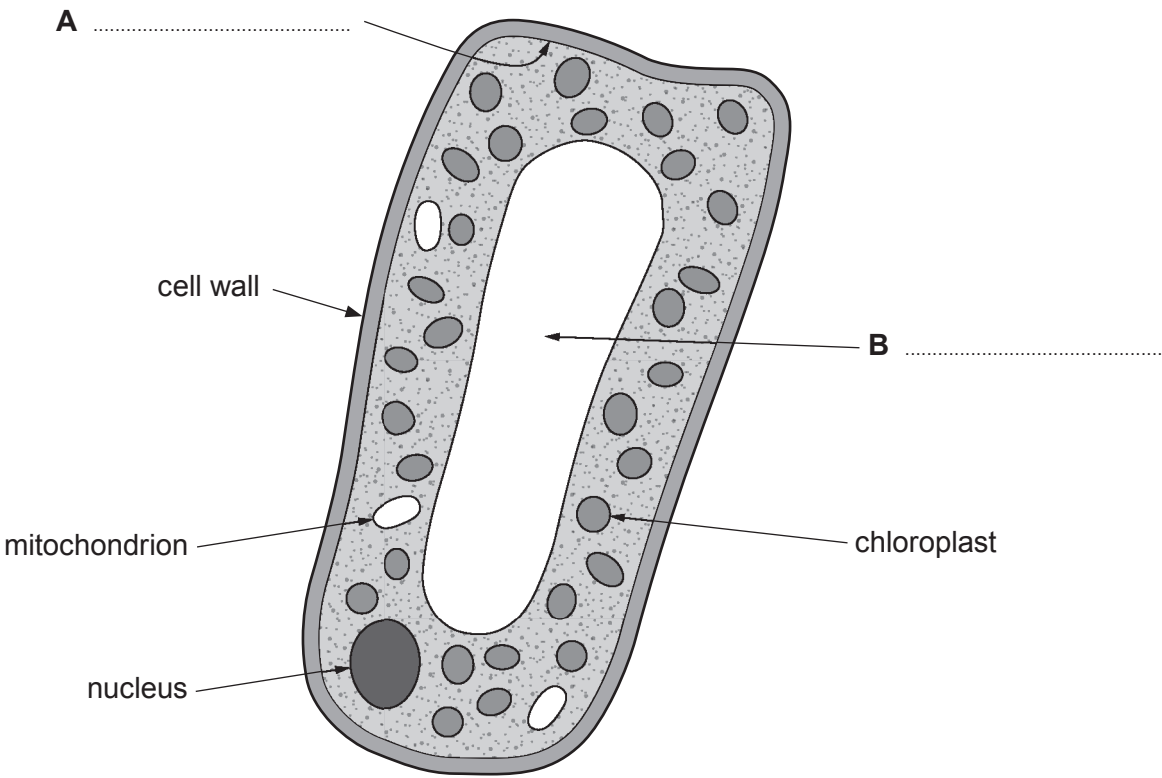
INFORMATION FOR CANDIDATES

The number of marks is given in brackets at the end of each question or part-question.
Question 7 is a quality of extended response (QER) question where your writing skills will be assessed.

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Answer **all** questions.

1. The diagram below shows a section through a plant cell as seen with a light microscope.

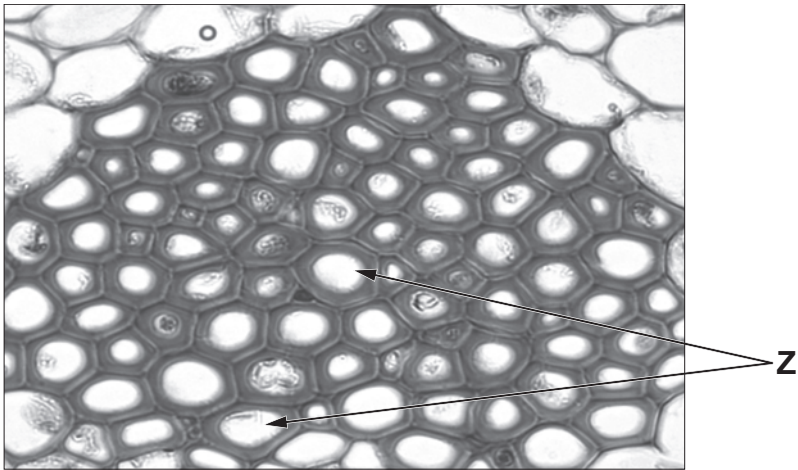


- (a) (i) Label parts **A** and **B** on the diagram. [2]
- (ii) Complete the table below. [3]

Part of cell	Function	Cell part present (✓) or absent (×) in animal cell
.....	photosynthesis	
mitochondrion		✓
.....	contains the chromosomes	

(b) The photograph below shows some plant cells as seen under a light microscope.

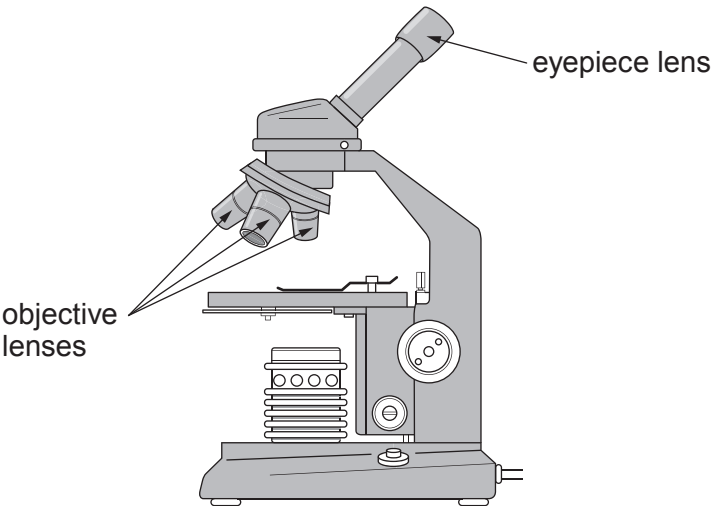
The group of cells labelled **Z** has been treated using a procedure to make their cell walls more clearly visible.



(i) What was the procedure carried out on these plant cells while they were prepared for viewing under the microscope? [1]

.....

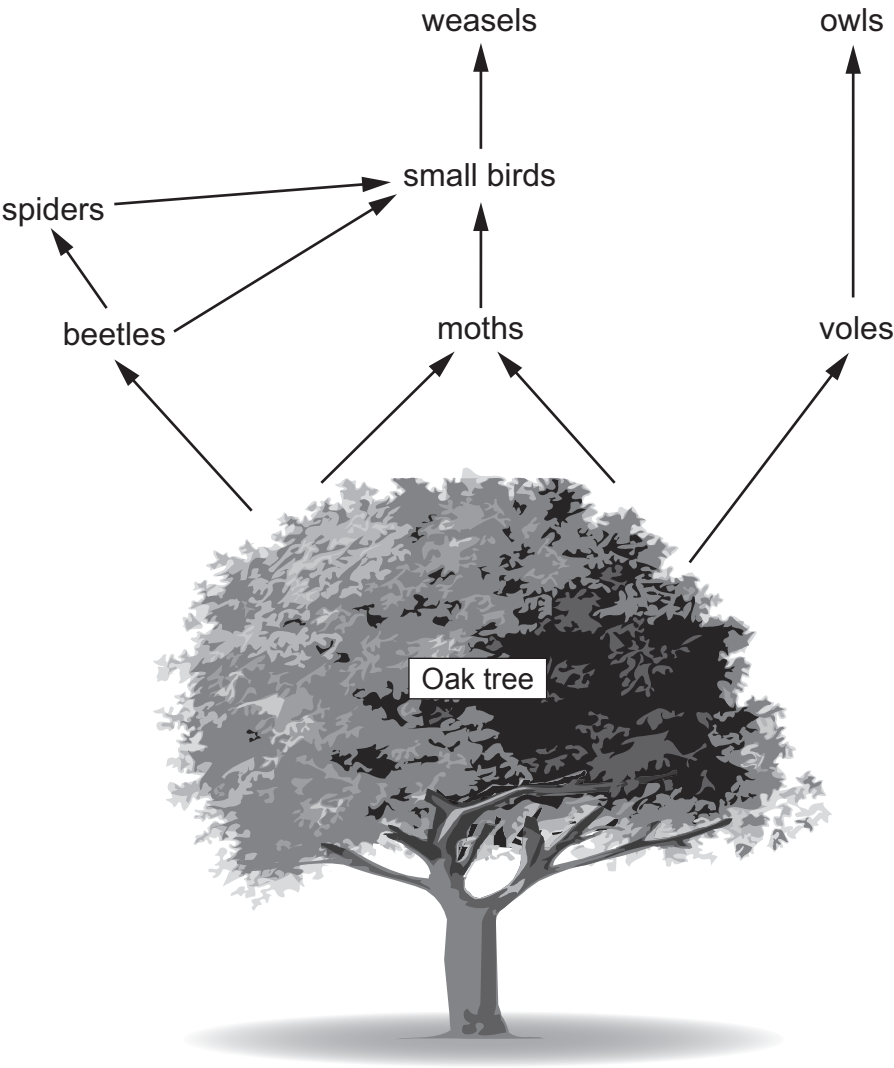
(ii) When the microscope was used to view these plant cells the magnification of the eyepiece lens was $\times 10$ and the magnification of the objective lens used was $\times 10$.



Using this information, calculate the total magnification of the image. [1]

Magnification = $\times$

2. The diagram below shows a food web.



(a) (i) State the source of energy for the food web. [1]

(ii) State **one** way in which energy can be lost from the food web. [1]

(b) Circle *true* or *false* in the table for each of the following statements about this food web. [3]

	Statement		
1	Moths and voles are first stage consumers.	true	false
2	Owls are third stage consumers.	true	false
3	Small birds are both third stage and second stage consumers.	true	false
4	Weasels and spiders are second stage consumers.	true	false
5	The oak tree is the only producer.	true	false

(c) (i) Complete the flow diagram below to show **one food chain** from the food web. [1]



(ii) **Draw a pyramid of biomass** to represent this food chain, in the space below.
Label each level in the pyramid with the name of the organism. [2]

3. Read the information about seagrass and use it to answer the questions.



Seagrass growing on the sea bed

Seagrass is a green plant which lives in the sea. It covers large areas of the sea bed.

Human activities such as dredging, building and industrial development can destroy seagrass. In the 1950s the loss of seagrass was 1.5 % per year but since 1990 scientists have calculated the loss to be 7 % per year.

Fisheries need sea grass. In the Mediterranean Sea, between 30 % and 40 % of the fish caught feed on seagrass and in Indonesian seas this is 60 %.

Seagrass is important in the carbon cycle, taking in carbon dioxide to produce glucose. This is converted into another carbon compound which can be stored in its large roots. Carbon dioxide makes sea water acidic which is harmful to sea organisms.

- (a) Answer the following questions using only the information given.

- (i) Give **one** reason why areas of seagrass are lost. [1]

- (ii) Which of the following statements describes the rate at which seagrass has been lost since 1950? [1]

- A** It has been constant every year
B It has increased
C It has decreased
D It has increased in some years and decreased in others.

Answer

(b) Answer the following questions using only the information given.

(i) Which statement describes how much the fishing industry depends on seagrass? [1]

- A Mediterranean and Indonesian fisheries depend totally on seagrass.
- B Mediterranean fisheries depend on seagrass for most of their fish.
- C Indonesian fisheries depend on seagrass for less than half of their fish.
- D Indonesian and Mediterranean fisheries depend on seagrass for some of their fish.

Answer

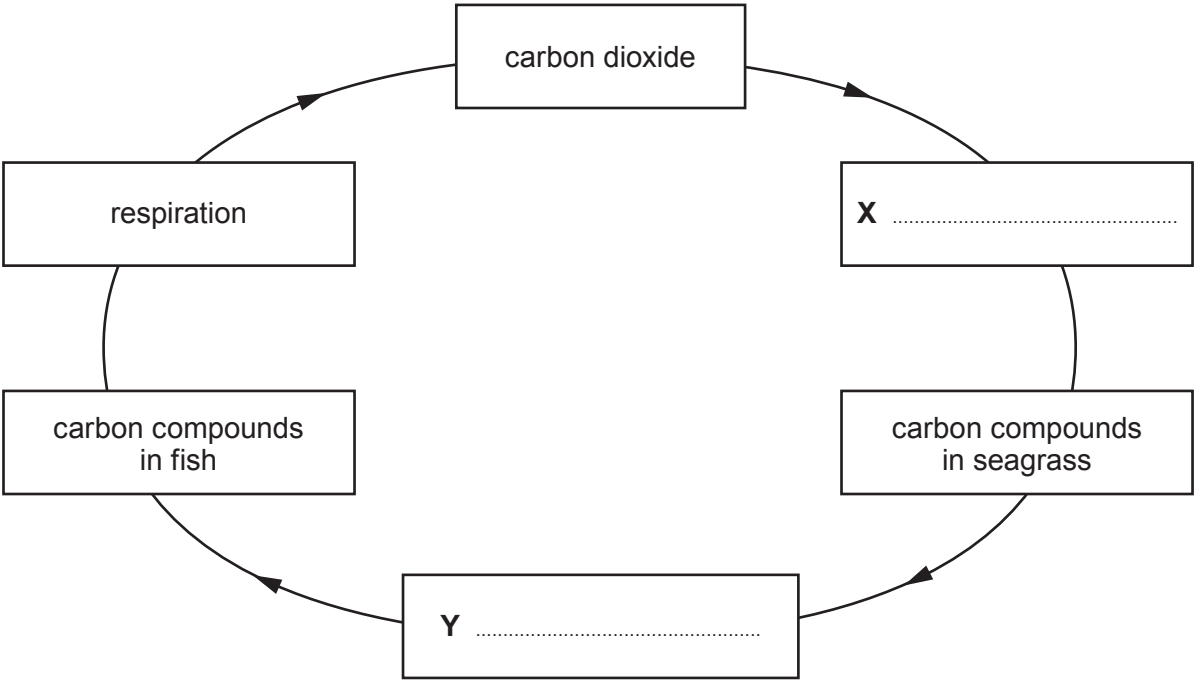
(ii) Apart from providing food, explain how seagrass is helpful to other species which live in the sea. [2]

.....

.....

(c) Answer the following questions using the information given and your own knowledge.

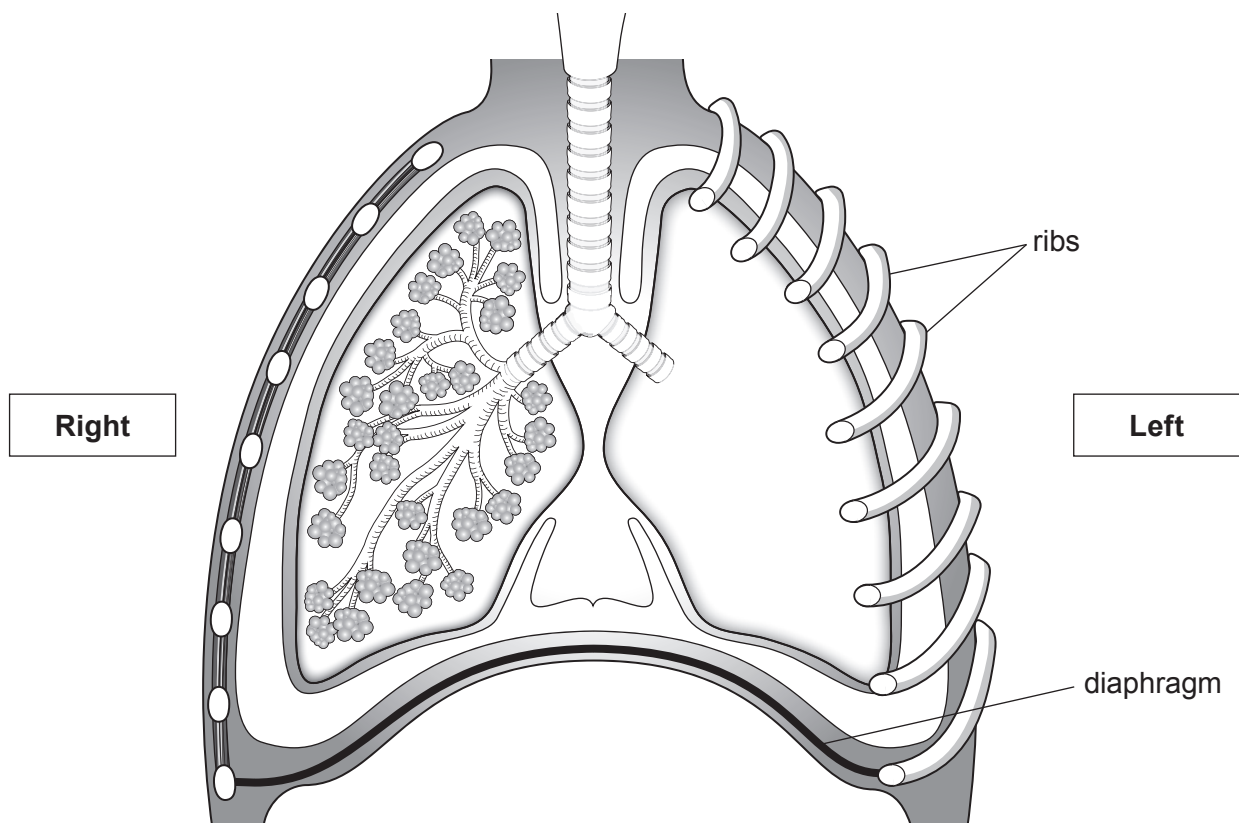
(i) Complete the diagram below which shows part of the carbon cycle, by naming processes X and Y. [2]



(ii) Name the carbon compound stored in the roots of seagrass. [1]

.....

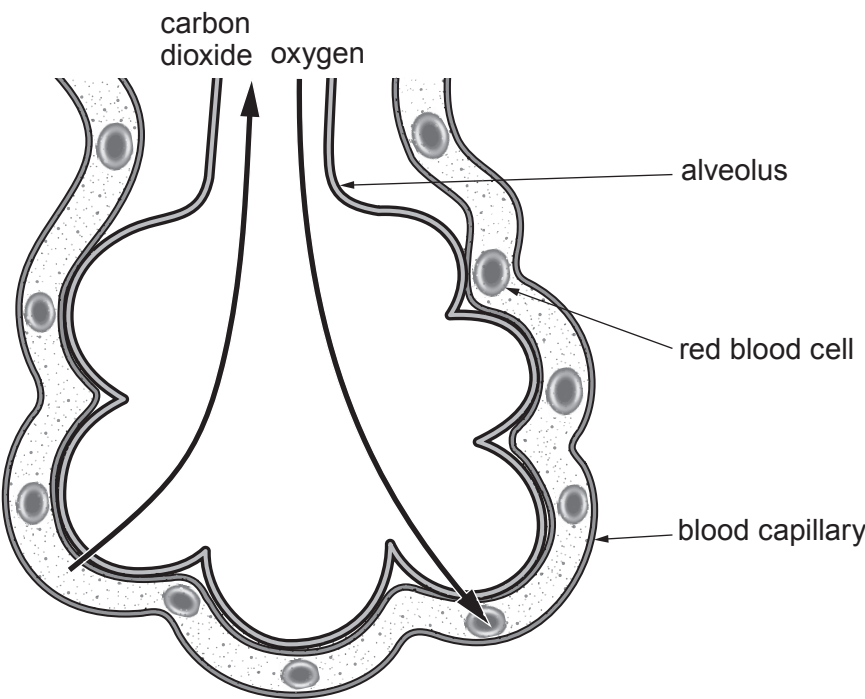
4. (a) The diagram below is a section through the human thorax during expiration (breathing out).



On the diagram, **draw**

- (i) **a line** to show the shape of the diaphragm at the end of inspiration (breathing in); [1]
- (ii) **an arrow** on the diagram to show the path taken by air passing from the trachea to a bronchiole in the **right** lung. [1]

(b) The diagram below shows a single alveolus.



Gas exchange between the alveolus and the blood in the capillary occurs through the wall of the alveolus.

Answer the following questions using information from the alveolus diagram above and your own knowledge.

(i) Name the process by which gas exchange occurs. [1]

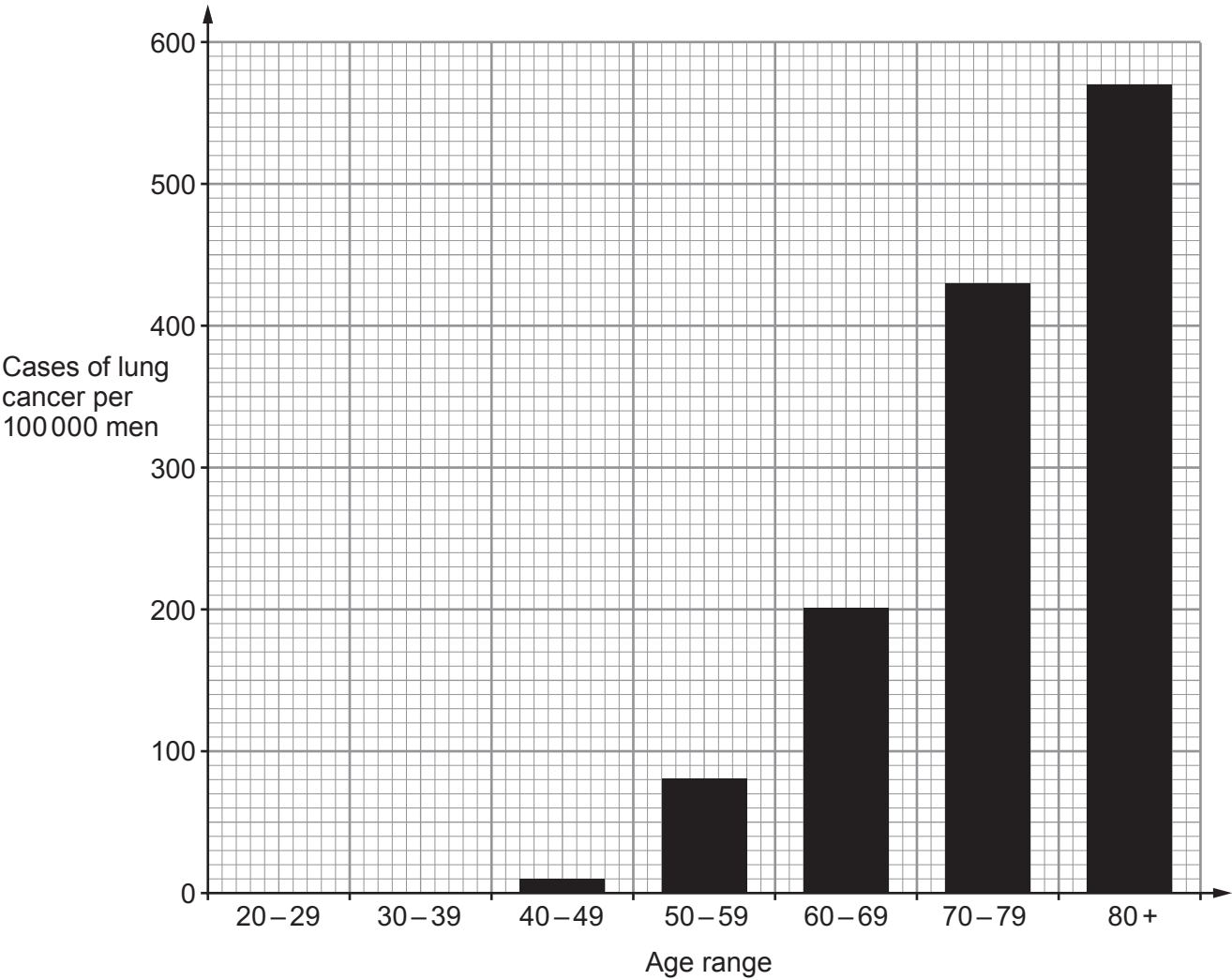
.....

(ii) Describe **two** ways in which the structure of the alveolus is adapted for this function. [2]

1.

2.

- (c) Cigarette smoking damages the lungs and causes lung cancer. In 2009, doctors investigated lung cancer and cigarette smoking in men in a city in the UK. The highest proportion of smokers was found to be in the age range 20-29. The cases of lung cancer are shown in the graph below.



- (i) State the conclusion the doctors should reach using the data given in the graph. [1]

- (ii) What is the evidence that lung cancer takes a long time to develop? [1]

.....

.....

.....

.....

Examiner
only

(iii) State **two** ways in which this investigation could be improved so that the doctors could be more confident that their results were representative of the whole of the population of the UK. [2]

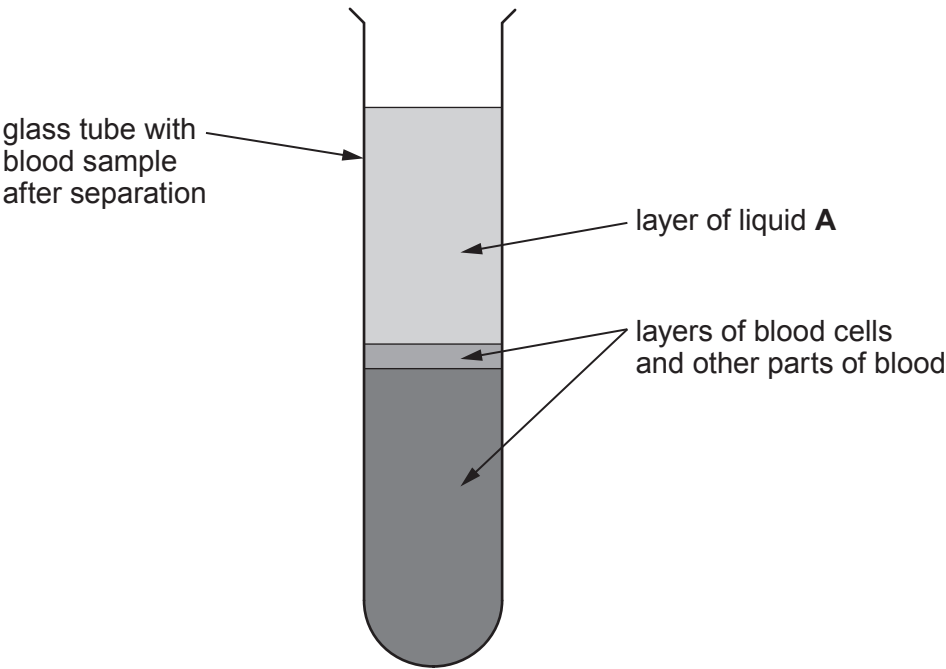
- 1.
- 2.

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5. A sample of human blood was placed in a test tube and the contents were separated by spinning the tube at high speed in a laboratory centrifuge. The diagram below shows the results.

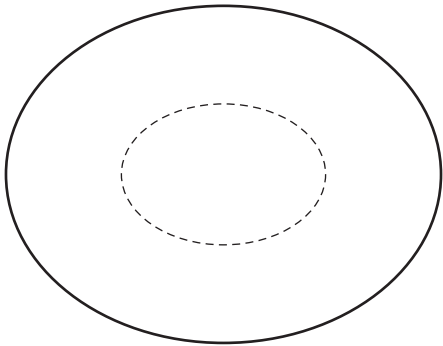


- (a) (i) The drawing below shows a red blood cell.

In the space, draw **another** type of cell which would be found in the sample.

State the name of the cell and its function.

[3]

Red blood cell  Function – carries oxygen	Name of cell Function
--	---

Examiner
only

- (ii) Name the liquid in layer **A** in the sample and state **two** substances which it transports. [3]

Name of liquid

Substances transported

1.

2.

- (iii) The volume of the blood sample was 20 cm³. Liquid **A** accounts for 45 % of this volume.
Calculate the volume of liquid **A**. [2]

Volume = cm³

- (b) A laboratory technician was working on some blood samples. She noticed that if the samples were left to stand for five minutes before placing them in the centrifuge, they could not be separated properly, as they had started to solidify.

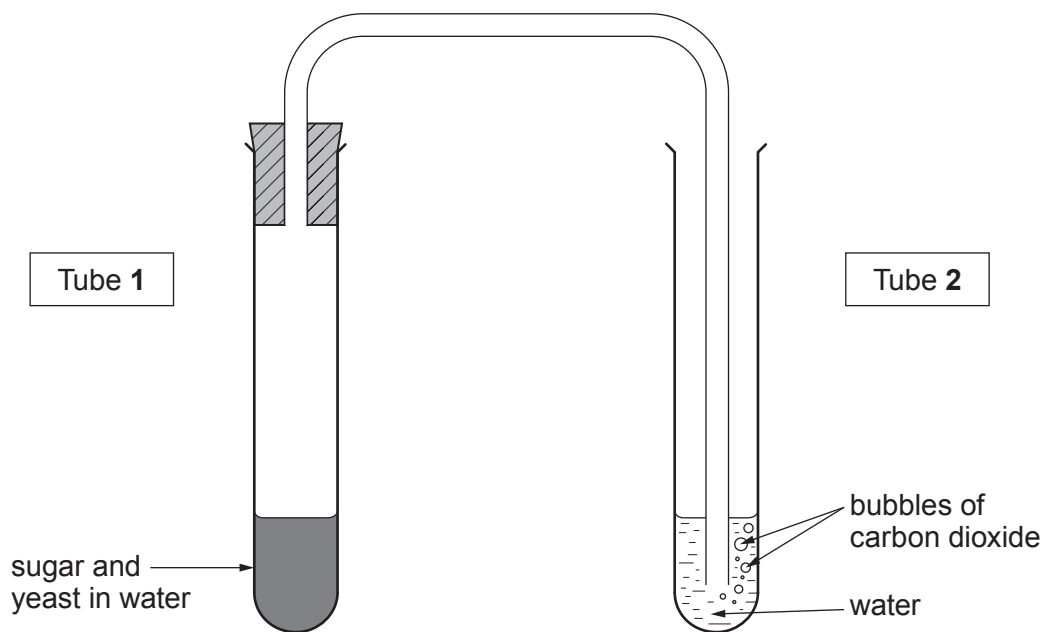
Use your knowledge of the functions of the parts of blood to explain this observation. [2]

.....
.....
.....

6. Respiration is a series of enzyme controlled reactions which release energy. The word equation for aerobic respiration is shown below.



Bethan investigated the effects of temperature on the rate of aerobic respiration in yeast. She used the apparatus below.



Bethan carried out the investigation at a range of temperatures from 15 °C to 45 °C.

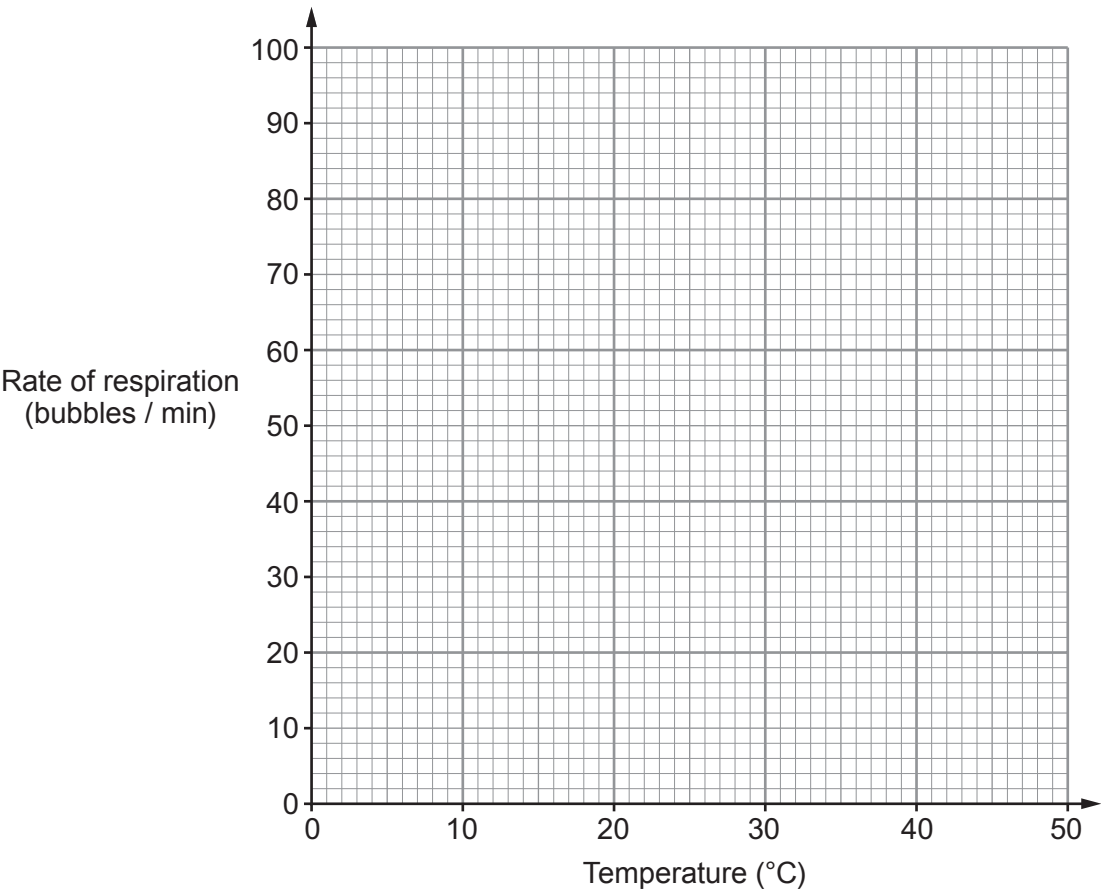
- (a) Name the apparatus she could use in order to control the temperature.

[1]

.....

(b) The table shows the results from the investigation.

Temperature (°C)	Rate of respiration (bubbles / minute)
15	19
25	50
30	64
35	80
40	84
45	67



- (i) Produce a graph of the results by:
- I. Plotting the values from the table.

II. Drawing a line to join your plots using a ruler.
- [3]

Examiner
only

(ii) I. Describe how the rate of respiration changes as the temperature is increased. [2]

.....

.....

II. Explain the reason for the change in the rate between 25 °C and 35 °C. [2]

.....

.....

(c) Bethan was told that she could improve her investigation by confirming that the bubbles of gas given off by the yeast actually were carbon dioxide.

(i) Name the solution she could use in tube **2** of the apparatus to test for carbon dioxide and state the positive result. [2]

.....

.....

(ii) State **two** factors which she should have kept constant in tube **1** throughout the investigation to ensure that it was a fair test. [2]

1.

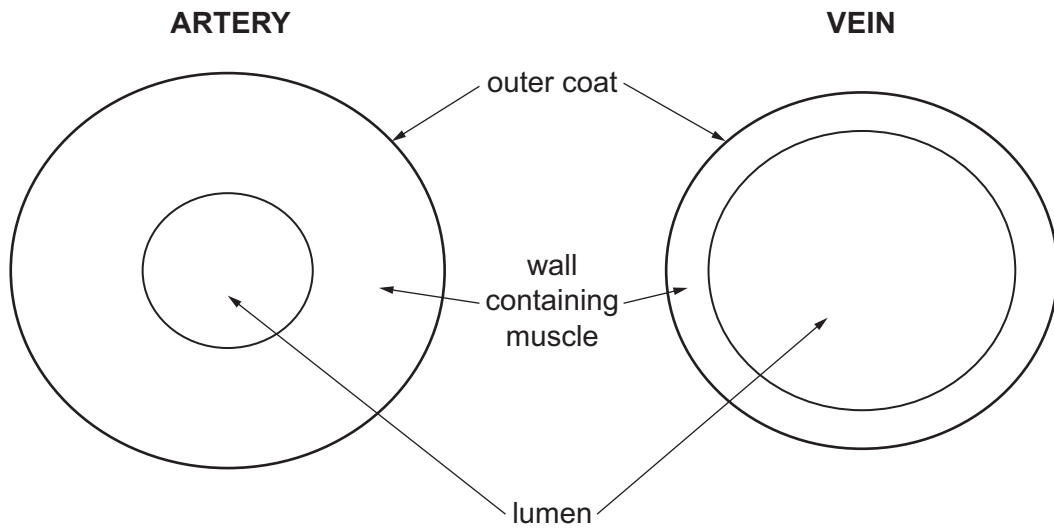
2.

12

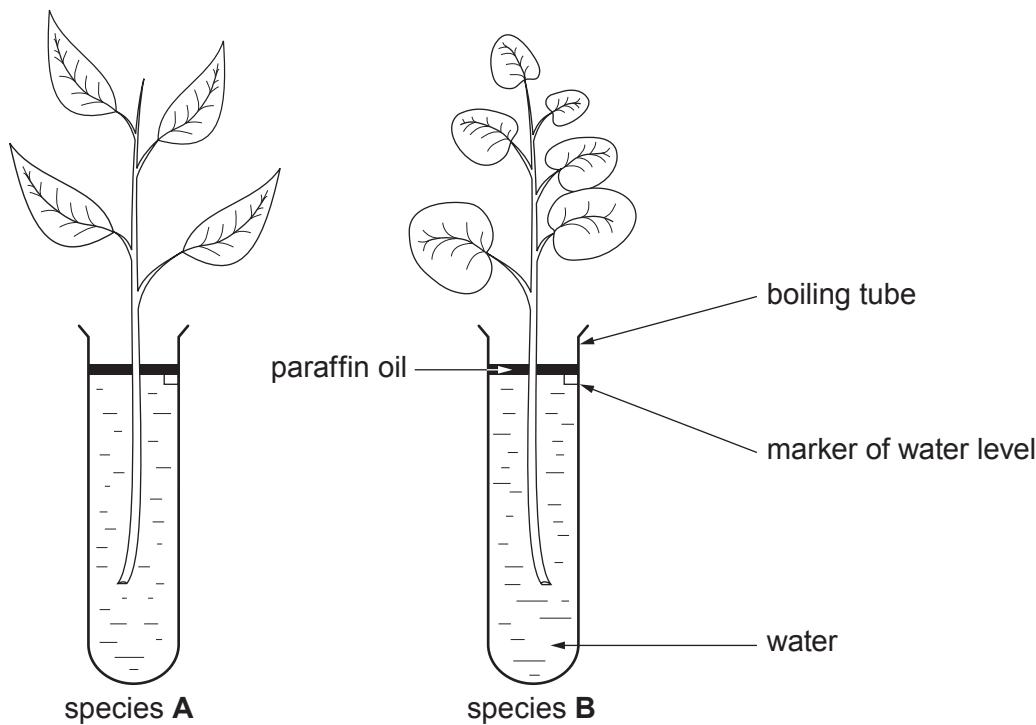
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- 7.** The diagrams show cross sections of an artery and a vein.

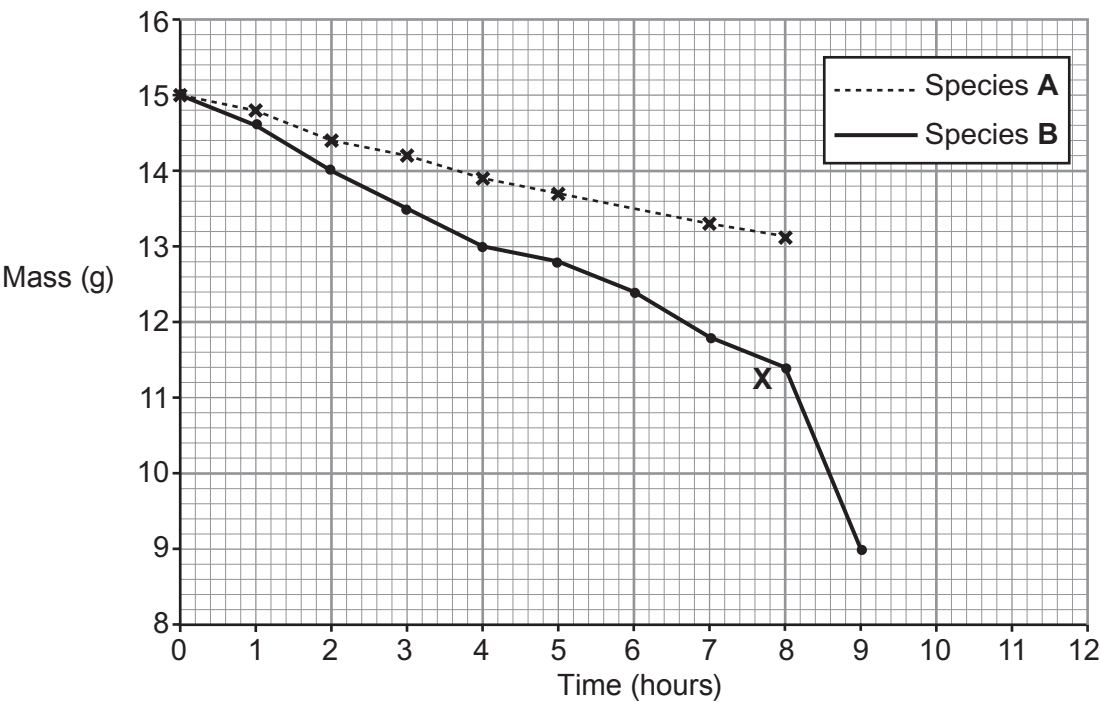
Compare the structure of arteries and veins and explain how they are related to their functions in the circulation of blood in the human body. [6 QER]



8. Shoots from two different species of plants, **A** and **B**, were placed in water in boiling tubes as shown below.



Both species were kept in the same conditions and their mass recorded at hourly intervals for eight hours. The results were recorded as line graphs shown below.



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- (a) (i) Use the data to calculate the difference in the loss in mass between species **A** and **B** at 8 hours. [1]

Difference in loss of mass = g

- (ii) State the term given to the process responsible for this loss of mass in plants. [1]

- (iii) Suggest **two** reasons for the difference in loss of mass between the two different species, **A** and **B**. [2]

1.

2.

- (iv) State the purpose of the paraffin oil in the boiling tubes. [1]

.....
.....

- (b) Which **one** of the following treatments was applied to species **B**, at point **X** on the graph, to produce the result shown at 9 hours?

Underline the correct answer. [1]

Enclosed in a bell jar

Upper surface of leaves coated with Vaseline

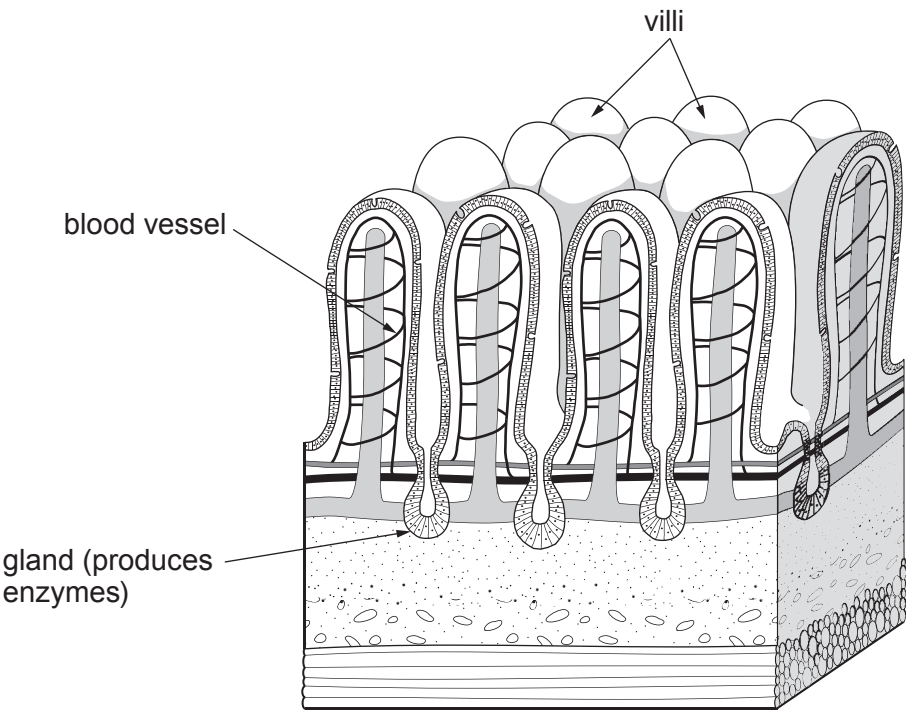
Half the leaves removed

Air directed at the leaves by a fan

- (c) What would you expect to happen to the rate of loss of mass if species **A** and **B** had roots? Give a reason for your answer. [2]

.....
.....
.....

9. The drawing shows the lining of part of the digestive system.



(a) Name the part of the digestive system that has this lining. [1]

(b) If the surface area of a single villus is 5 mm^2 , calculate the total surface area of all the villi shown. Assume that all the villi are complete and include a unit in your answer. [1]

Total surface area =
unit =

(c) State **two** functions carried out in this part of the digestive system. [2]

Function 1.
Function 2.

(d) State **one** feature of the digestive system which helps this part carry out each function named in (c). [2]

Function 1.

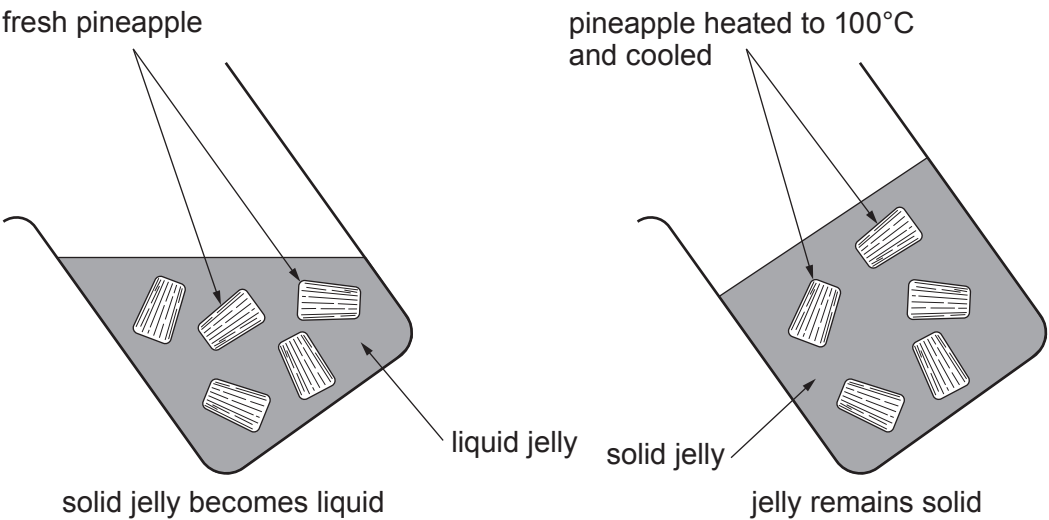
Function 2.

(e) Fresh pineapple is sometimes added to meat to make it tender. Fresh pineapple contains a protease enzyme.

Alan investigated how this took place. He used the following:

- fresh pineapple
- pineapple that had been heated to 100 °C and cooled
- a solid jelly made from protein called gelatin

Two samples of solid jelly were made and the pineapple was pushed into them. After 24 hours the results were shown below:



Explain the results with:

(i) fresh pineapple; [3]

.....
.....
.....
.....
.....

(ii) pineapple heated to 100°C and cooled. [3]

.....
.....
.....

Surname	Centre Number	Candidate Number
Other Names		0



GCSE – **NEW**

3400U10-1



BIOLOGY – Unit 1:
Cells, Organ Systems and Ecosystems

FOUNDATION TIER

MONDAY, 11 JUNE 2018 – MORNING

1 hour 45 minutes

For Examiner's use only		
Question	Maximum Mark	Mark Awarded
1.	5	
2.	9	
3.	6	
4.	7	
5.	5	
6.	9	
7.	11	
8.	8	
9.	10	
10.	10	
Total	80	

ADDITIONAL MATERIALS

In addition to this paper you may require a calculator and a ruler.

INSTRUCTIONS TO CANDIDATES

Use black ink or black ball-point pen. Do not use gel pen. Do not use correction fluid.
Write your name, centre number and candidate number in the spaces at the top of this page.
Answer **all** questions.
Write your answers in the spaces provided in this booklet. If you run out of space, use the additional page at the back of the booklet, taking care to number the question(s) correctly.

INFORMATION FOR CANDIDATES

The number of marks is given in brackets at the end of each question or part-question.
Question 8(a) is a quality of extended response (QER) question where your writing skills will be assessed.



JUN183400U10101

Answer all questions.

1.

What is a healthy diet?



In 2015, the UK Food Standards Agency and the Department of Health provided advice on which food should be eaten to maintain good health. This is summarised below.

The energy content of food should be balanced weekly according to a person's activity over the seven days.

To ensure a healthy diet everyone should have:

- *a high proportion of foods containing starch*
- *plenty of fruits and vegetables as these are high in fibre and useful micronutrients such as vitamins and minerals*
- *more fish as it contains protein and omega 3 oils*
- *fewer 'ready meals' to avoid high levels of sugar, saturated fat and salt as well as excess energy content*
- *less processed meat such as sausages, which do contain protein but also high levels of salt*
- *plenty of water and also skimmed milk, which contains vitamins and is low in fat*



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Use the information opposite and your own knowledge to answer the following questions.

(a) Based on this information, write 'true' or 'false' for each of the following statements. [3]

	Statement	True or False
1	All fats and oils should be avoided.	
2	All micronutrients come from fruits and vegetables.	
3	Fruits and vegetables provide bulk during digestion.	
4	It is better to obtain protein from fish than from processed meat.	
5	The energy in the diet needs to be balanced every day.	

(b) Eating large numbers of 'ready meals', can lead to ill health in humans. State **one** condition that can arise from eating high levels of:

(i) sugar [1]

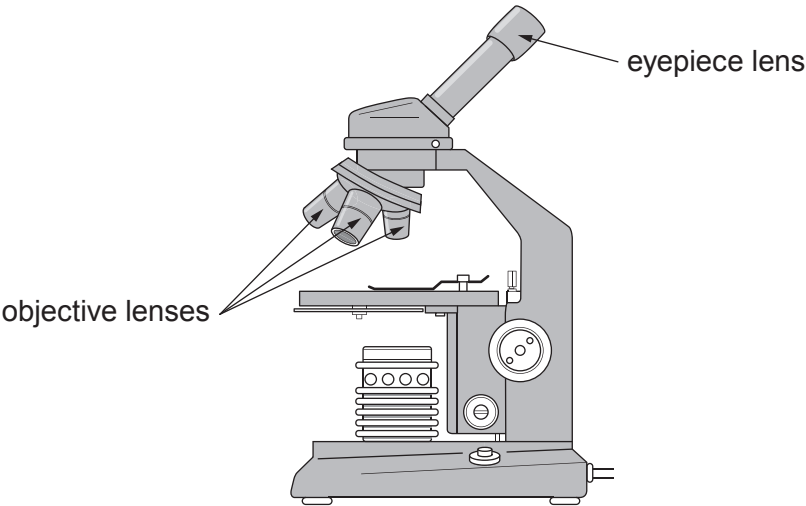
(ii) salt [1]

5

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03



2. Rheinallt observed some plant cells using a light microscope.



(a) He set up the microscope so that it had a total magnification of $\times 240$.

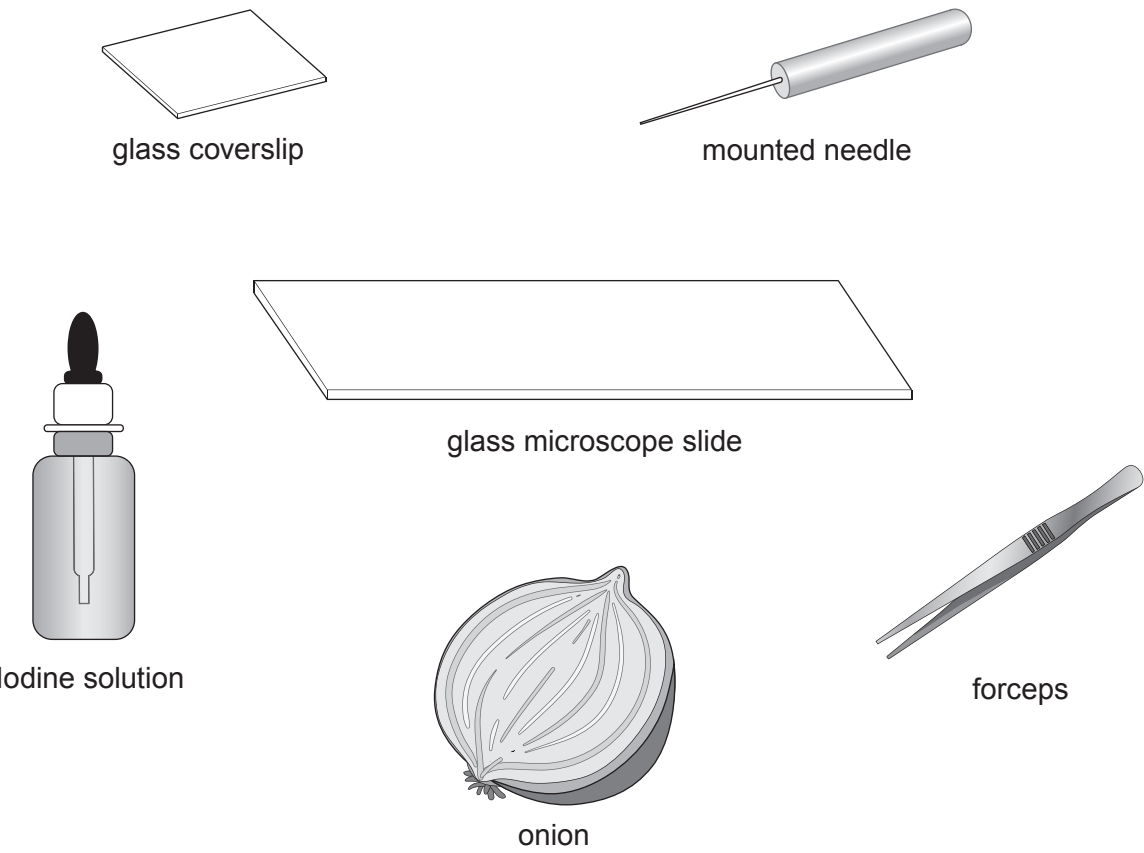
The magnification of the eyepiece lens on his microscope was $\times 6$. Which of the following objective lenses did he use? [1]

Objective lens power	Magnification
low	$\times 8$
medium	$\times 15$
high	$\times 40$

Objective lens magnification = $\times$



(b) The diagram shows the apparatus Rheinallt used to prepare a slide of onion cells.



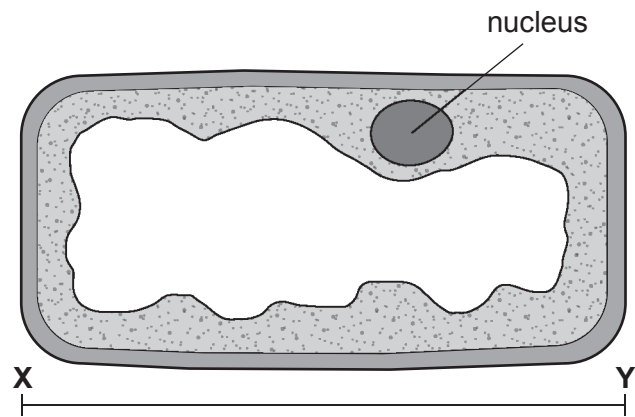
Continue and complete the method below, which he used to prepare his slide for observation under the microscope. [3]

Method

- Take a piece of freshly cut onion.
- Use forceps to carefully peel a thin layer of cells from the inner surface of the onion.
- Place the layer of onion cells onto the centre of a microscope slide.
-
-
-



(c) Rheinallt made a large drawing of one of the cells he observed.



- (i) I. Measure the length of the cell in Rheinallt's drawing along line **X–Y**. [1]

Length of cell in drawing = mm

- II. The actual length of cells of this type is usually 0.02 mm.
Use your answer to part I. to calculate the magnification of Rheinallt's drawing. [2]

Magnification of drawing = ×

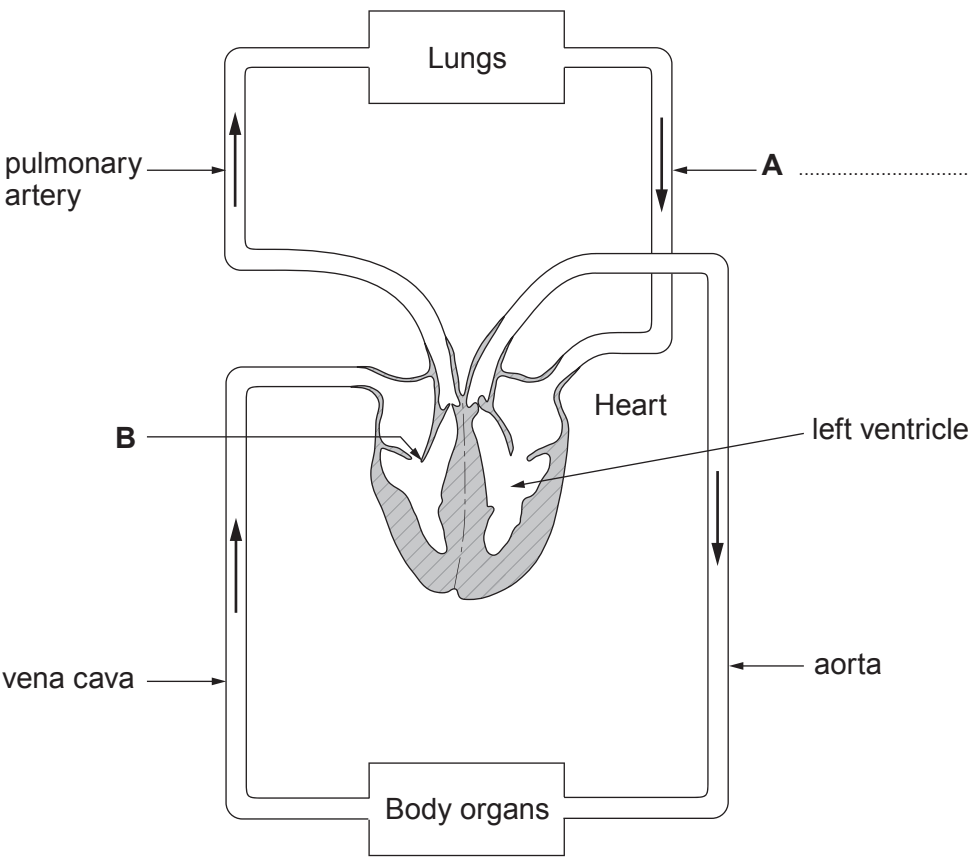
- (ii) From the drawing, give **one** feature of the cell which shows that it is a plant cell. [1]

.....

- (d) Rheinallt compared his drawing with an image obtained from an electron microscope.
Why would the electron microscope image give more information about the structure of an onion cell? [1]

.....

3. The diagram shows the human circulatory system.



- (a) (i) Label the blood vessel **A** on the diagram. [1]
- (ii) State the function of structure **B**. [1]

.....



(b) The table shows blood pressures in different parts of the circulatory system.

Blood vessel	Maximum blood pressure (kPa)
pulmonary artery	3.3
aorta	16.0
capillary in body organ	2.0
left ventricle	17.0
right ventricle	3.5

- (i) I. Calculate the difference in the maximum blood pressure between the aorta and the pulmonary artery. [1]

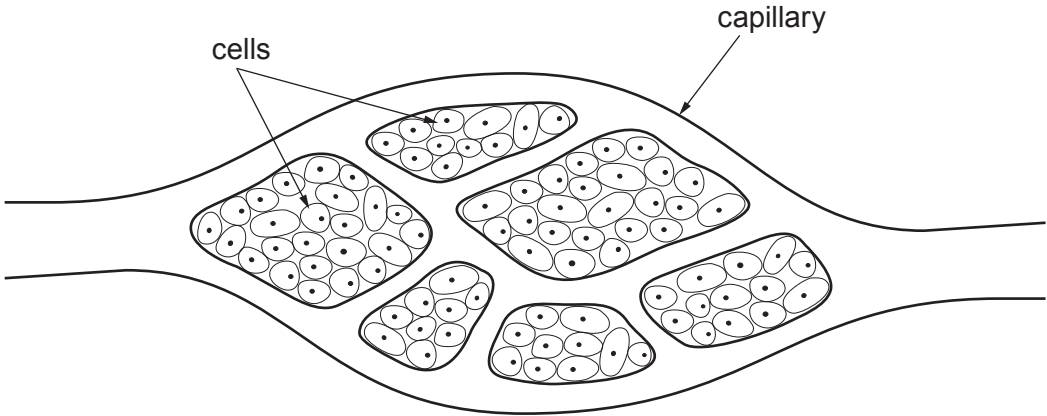
Difference in pressure = kPa

- II. State the reason for the difference in blood pressure in these two blood vessels. [1]

.....

.....

- (ii) The diagram shows some capillaries in an organ of the body.



Blood flows very slowly through the capillaries, allowing useful substances in the blood and waste products in cells to be exchanged.

- I. From the table above, what is the evidence that the blood flows slowly through the capillaries? [1]

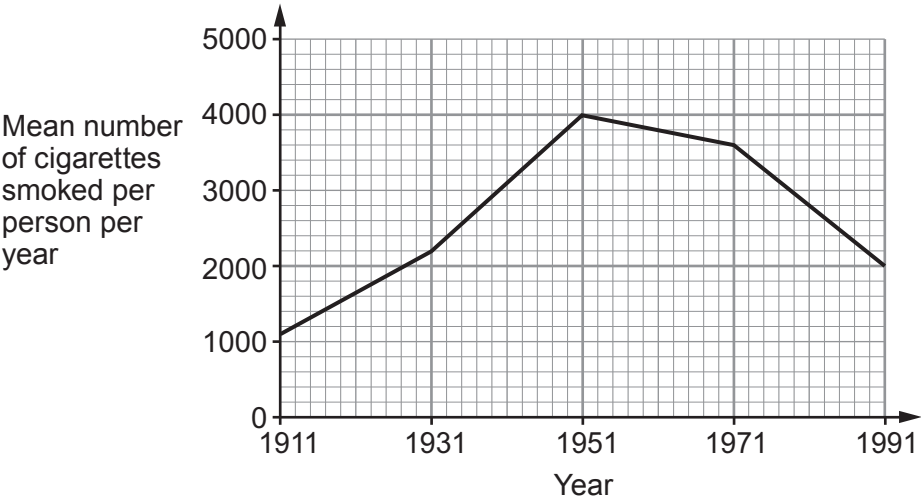
- II. State how the capillary walls are well adapted for the exchange of substances. [1]

.....



4. (a) Cigarettes contain chemicals which cause cancer.

The graph below shows the mean number of cigarettes smoked per person per year and the table shows the number of deaths caused by lung cancer in the UK over a period of 80 years.



Year	Deaths per year from lung cancer
1911	600
1931	1 500
1951	14 000
1971	22 000
1991	19 000

Describe what the data in the graph and the table show about the link between the mean number of cigarettes smoked per year and the deaths from lung cancer over the time period 1911-1991. [3]

.....

.....

.....

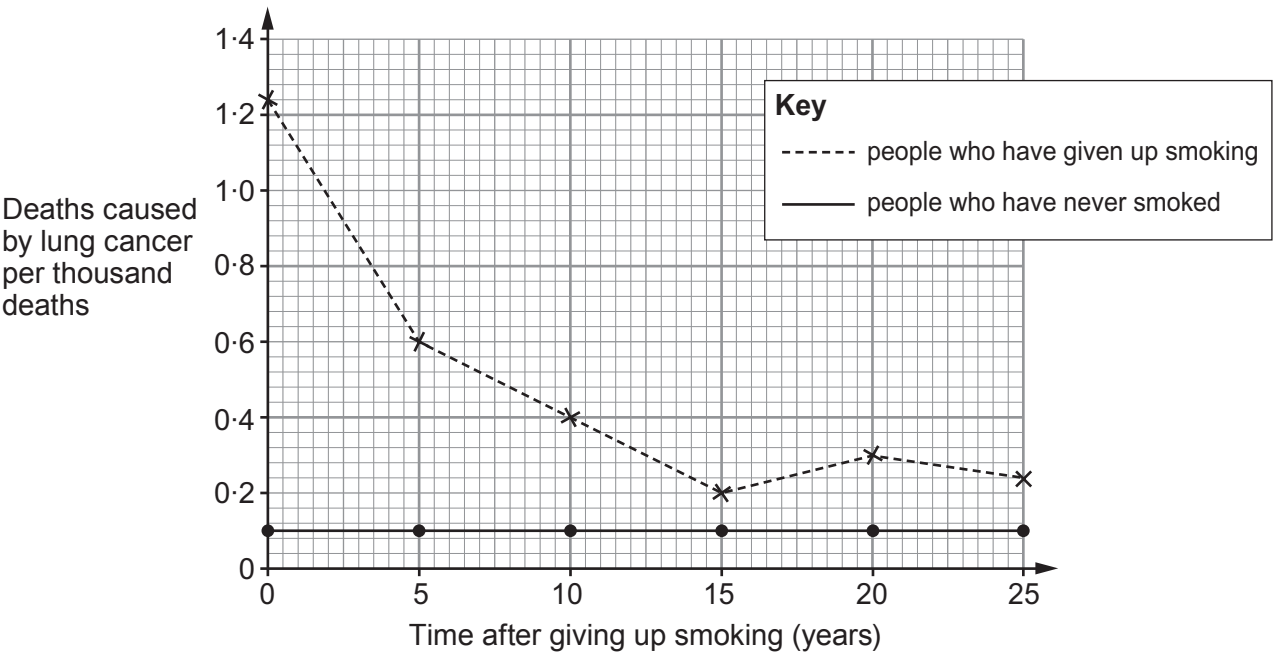
.....

.....

.....



(b) The graph shows the effect of giving up smoking on deaths from lung cancer.



Use the graph above to complete the sentences. Choose the correct answer.

(i) If a person gives up smoking, the risk of dying from lung cancer will be: [1]

- A the same as a non-smoker after 10 years.
- B twice the risk of a non-smoker after 15 years.
- C removed completely after 25 years.
- D unchanged even after 20 years.

Answer

(ii) The data in the graph show that: [1]

- A Cigarettes cause lung cancer.
- B Over 1200 smokers die from lung cancer every year.
- C It is possible for a non-smoker to die from lung cancer.
- D The number of people who do not smoke has increased in recent years.

Answer



(c) In an investigation of 600 cigarette smokers, almost half said they wanted to give up. At the end of one year only 42 had actually succeeded.

Explain why many people find it very difficult to give up smoking cigarettes. [2]

.....

.....

.....

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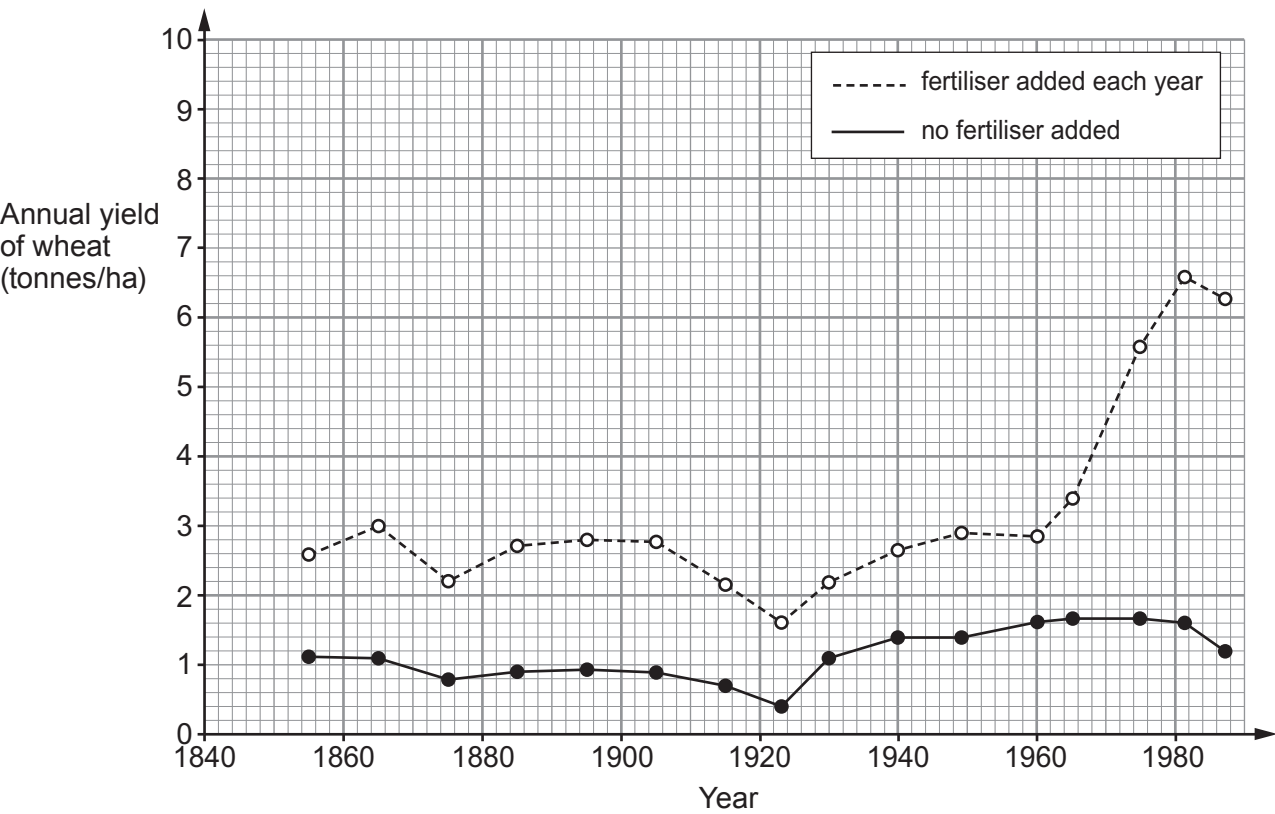


5. The information below is about the growth of wheat plants in a long-term investigation.



Wheat growth experiment at a research centre in Mid Wales

The graph shows the results of an investigation into the use of fertiliser. One field had the same mass of fertiliser added each year from 1855. Another field had no fertiliser added in that time.



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- (a)

(i)

From the graph, what evidence is there that:

I.

the addition of fertiliser increased the growth of wheat plants;

[1]

.....

II.

the fertiliser took a long time to reach its full effect?

[1]

.....

(ii)

How were the data from the field without fertiliser important to the scientists in interpreting their results?

[1]

.....
- (b)

Farmers now frequently add NPK fertiliser to their fields.
State the way in which the following mineral nutrients in NPK fertiliser affect the growth of their crop plants.

[2]

(i)

Effect of phosphate

.....

(ii)

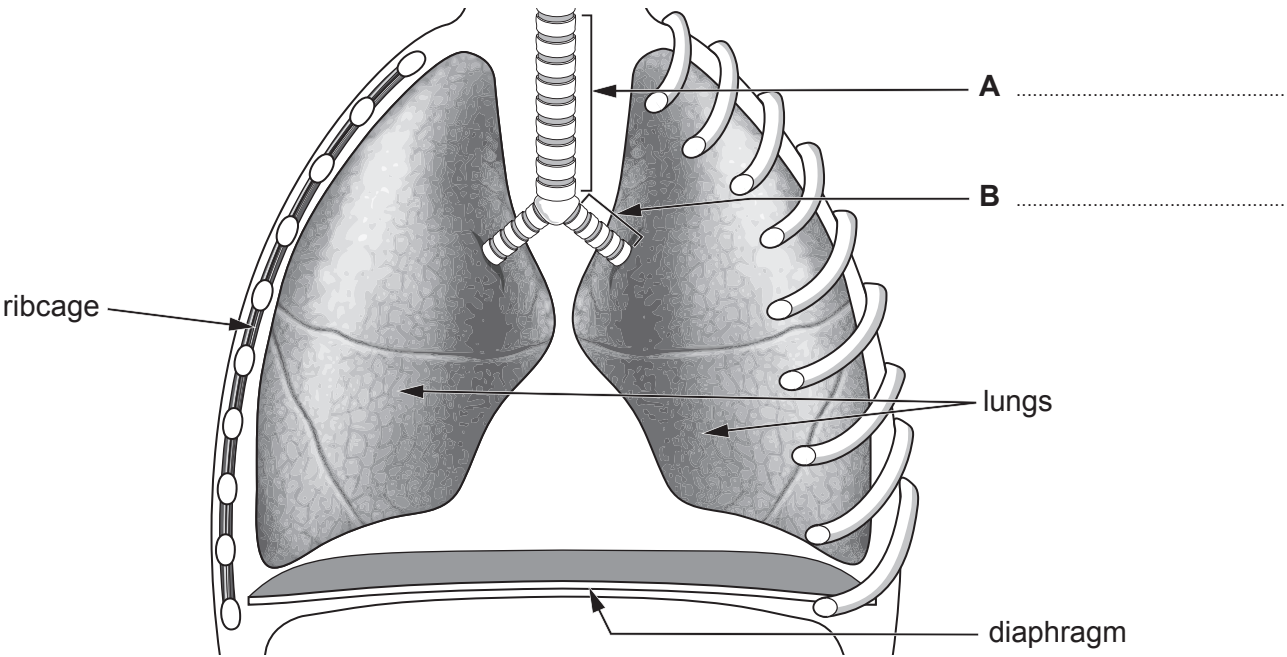
Effect of potassium

.....

5



6. The diagram shows a vertical section through the human thorax after inspiration.



- (a) (i) Label structures **A** and **B** on the diagram. [1]
- (ii) Describe the movement of the diaphragm during **expiration** and how this affects the volume and pressure in the chest cavity. [3]

.....

.....

.....

.....

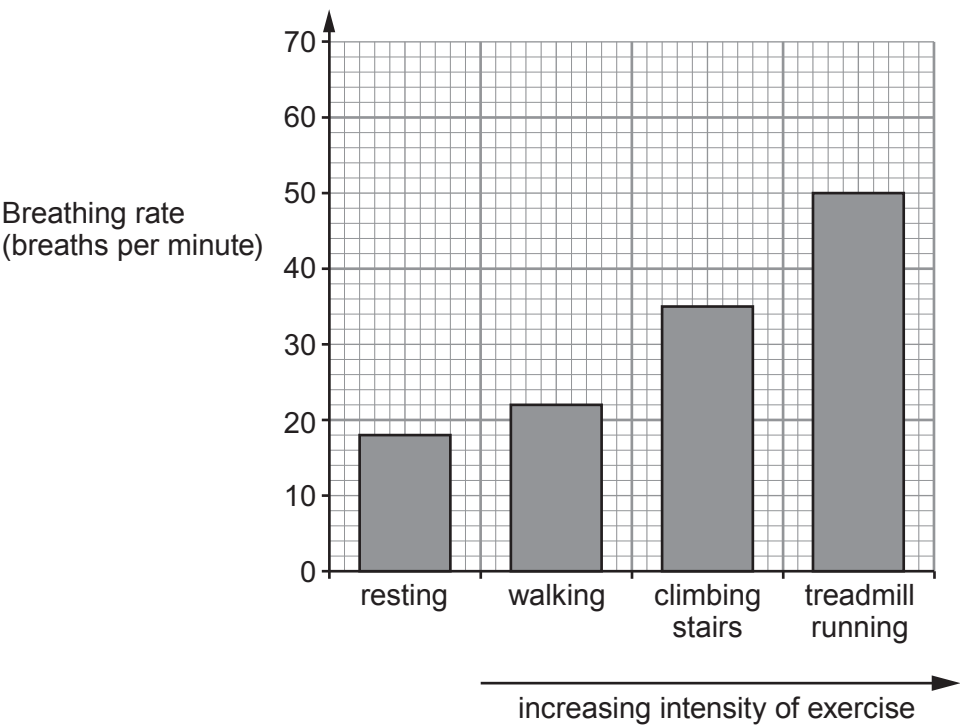
.....

.....



- (b) Two students investigated breathing rates (the number of breaths per minute) while at rest for five minutes and during three different activities, each of which lasted five minutes.

One student did the activities and the other recorded the results.



- (i) From these results, what could you conclude about the effect of exercise on the breathing rate? [2]

.....

.....

- (ii) State **one** way in which the students could have increased the confidence they had in their results. [1]

.....

- (c) Explain why the breathing rate changes when a person starts exercising. [2]

.....

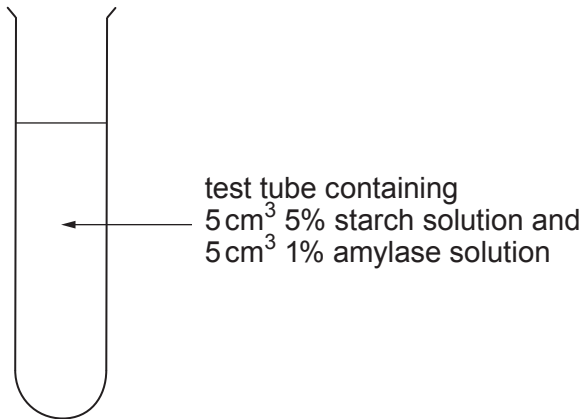
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7. Students investigated the activity of amylase enzyme from the bacterium *Bacillus licheniformis* which is used in industry. Amylase digests starch.

They set up six test tubes, each as shown in the diagram below, at a range of temperatures from 10–60°C and measured the amylase activity after 5 minutes.



(a) The students tried to ensure that their investigation was a fair test.

State **one other** variable which they should have kept constant in all the test tubes. [1]

The results of the investigation are shown in the table below.

Temperature (°C)	Amylase activity (a.u.)
10	0.3
20	0.9
30	1.4
35	1.6
40	1.8
60	0.4

(b) Draw a graph of the results on the grid opposite by

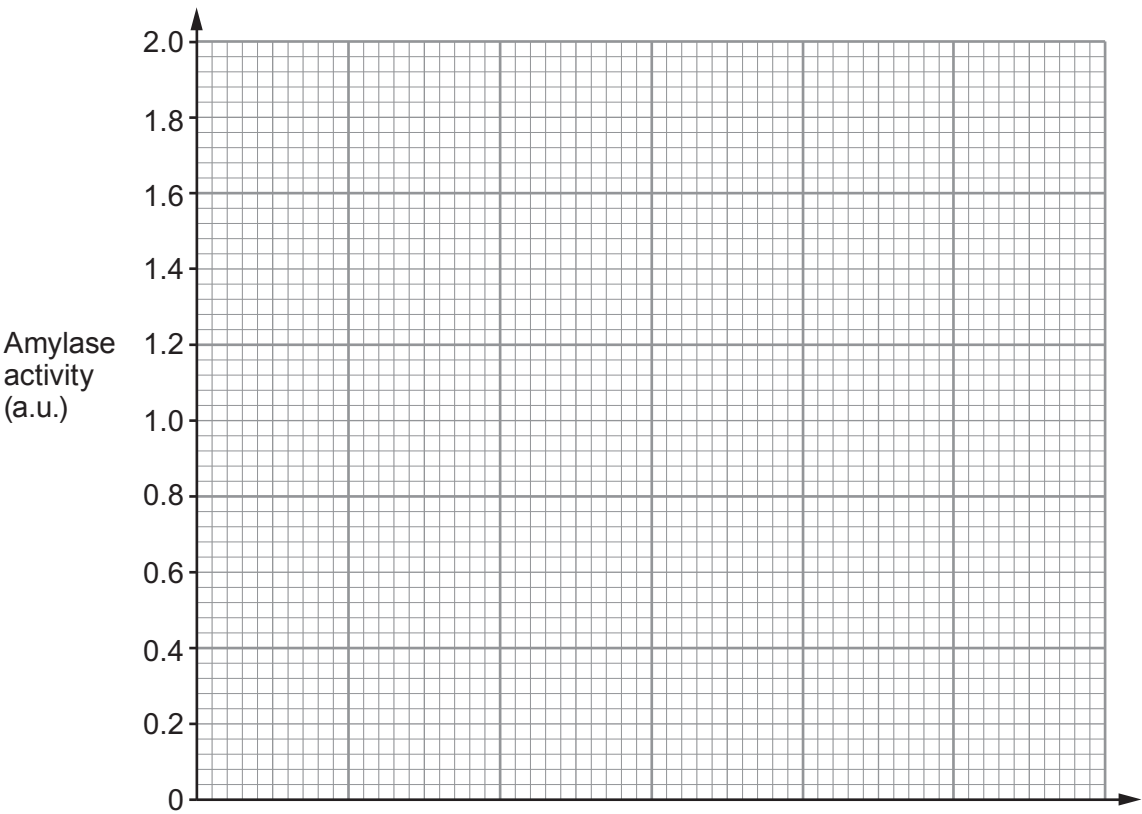
(i) Choosing a scale for temperature and labelling the axis. [1]

(ii) Plotting the points. [2]

(iii) Join your plots using a ruler. [1]



Examiner
only



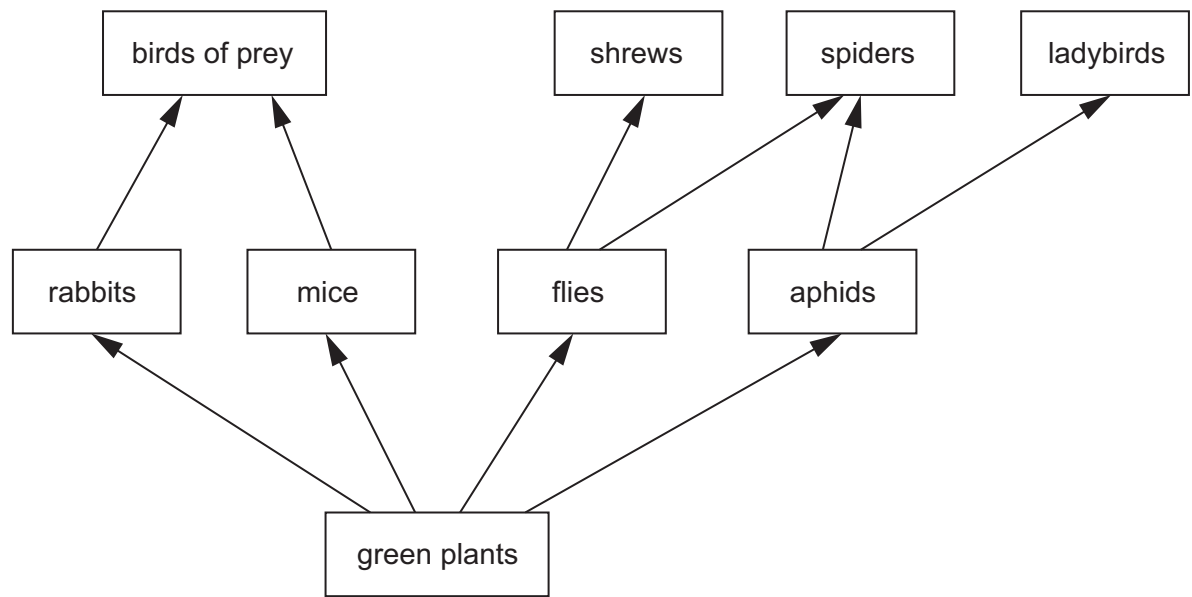
- (c) From your graph,
- (i) State how the amylase activity changes between 15°C and 35°C and explain the reason for this change. [2]
.....
.....
 - (ii) State the reason for the change in the amylase activity between 40°C and 60°C. [1]
.....
 - (iii) I. Use your graph to suggest the optimum temperature for this enzyme. [1]

Optimum temperature = °C

II. State why the optimum temperature cannot be identified with accuracy from these results. Suggest how the investigation could be improved to allow this temperature to be identified more accurately. [2]
.....
.....
.....



8. The diagram below shows a food web in an area of grassland.



Farmers were concerned that the rabbit population was increasing and called in a pest control company to destroy the rabbits.

(a) Identify the herbivores and second stage consumers in the food web above. Explain how the populations of each would be affected if all the rabbits were destroyed. [6 QER]

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Examiner
only

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(b) Explain how the levels of mineral nutrients in the soil would be increased if the dead bodies of rabbits were not removed by the pest control company. [2]

.....

.....

.....

8



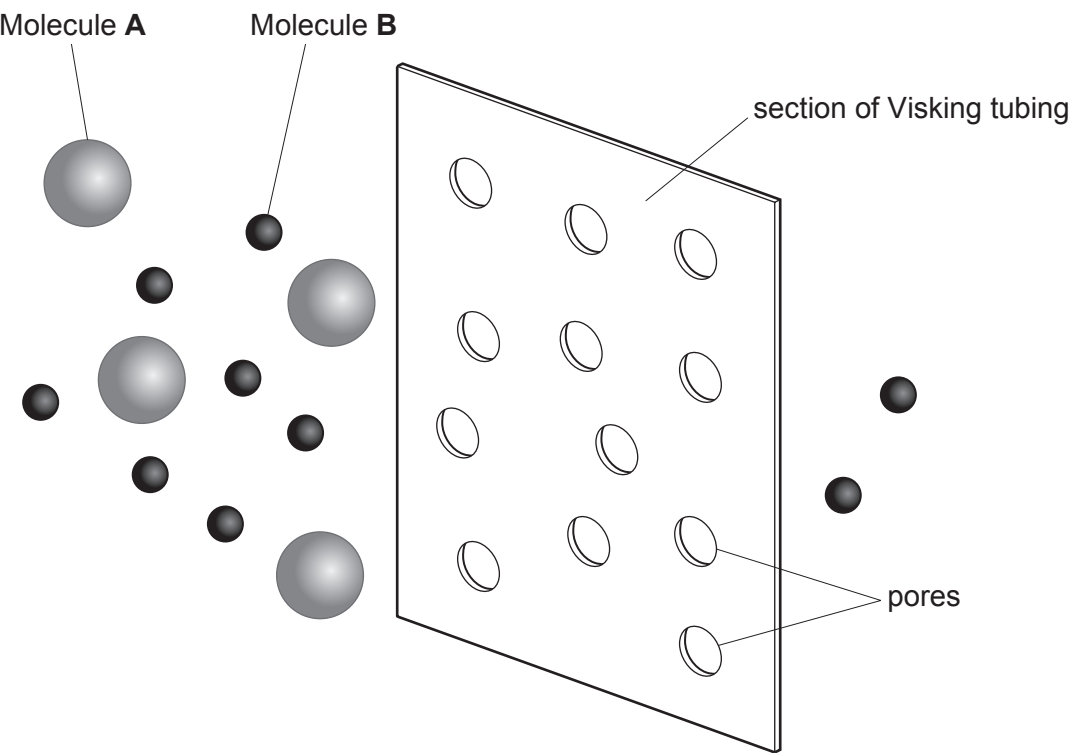
9. (a) What is meant by the term selectively permeable membrane? [1]

.....

.....

Visking tubing can be used as a model of the cell membrane.

The diagram below shows a section of Visking tubing working in the same way as a selectively permeable membrane.



Use the diagram above and your own knowledge to answer the following:

(b) (i) State the process by which molecules could pass through the Visking tubing. [1]

.....

(ii) Identify which molecules pass through the Visking tubing. Explain your answer. [2]

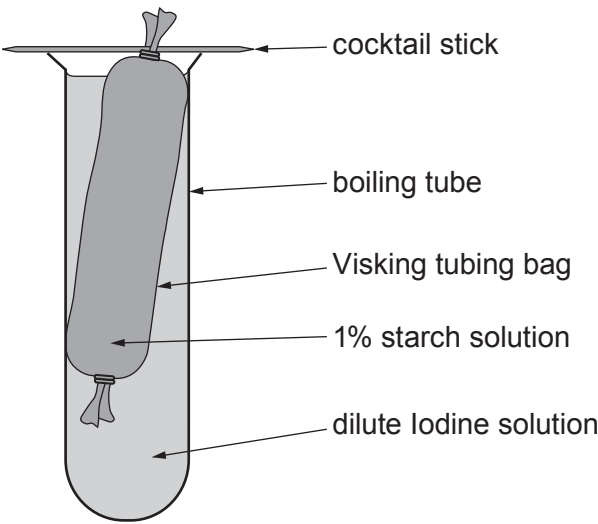
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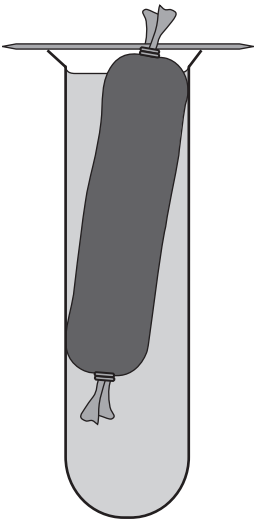
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(c) Students were instructed to set up the following apparatus.



The appearance of the apparatus after 15 minutes is shown below.



(i) Explain why the colour of the contents inside the Visking tubing turned blue black. [3]

.....

.....

.....



Examiner
only

(ii) Explain why the colour of the Iodine solution in the boiling tube did not change. [2]

.....

.....

.....

(d) Name **one** substance required for respiration that would pass into a cell. [1]

.....

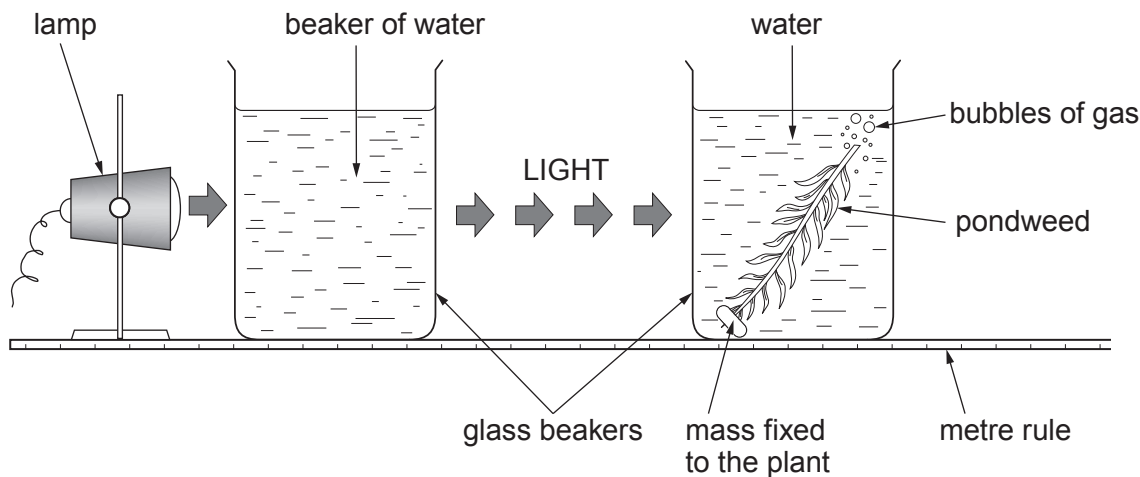
10



10. (a) (i) Write the **word** equation for photosynthesis. [2]

(ii) State the name of the pigment present in plant cells which absorbs light. [1]

Phoebe and Adam used the apparatus below to study the rate of photosynthesis in the pondweed (*Elodea sp.*).



The number of gas bubbles per minute produced by the pondweed was counted at different distances from the light.

The experiment was carried out three times at each distance.

The results are shown below. Means were calculated to the nearest whole number.

Distance of lamp from pondweed (cm)	Number of bubbles per minute			
	Test 1	Test 2	Test 3	Mean
10	19	32	25	25
20	14	20	20	18
30	11	15	17	
40	7	10	13	10
50	5	9	11	8



Examiner
only

(b) **Complete the table** opposite by calculating the mean number of bubbles for a distance of 30 cm. **Write your answer in the table.** [2]

(c) State the relationship between the distance of the lamp from the pondweed and number of bubbles produced per minute. Explain your answer. [3]

.....

.....

.....

.....

(d) Explain why a beaker of water was placed between the lamp and the pondweed. [1]

.....

.....

(e) State how you could improve the accuracy of this investigation. [1]

.....

END OF PAPER

10



Surname	Centre Number	Candidate Number
Other Names		0



GCSE

3400U10-1



BIOLOGY – Unit 1:
Cells, Organ Systems and Ecosystems

FOUNDATION TIER

FRIDAY, 7 JUNE 2019 – AFTERNOON
1 hour 45 minutes

For Examiner’s use only		
Question	Maximum Mark	Mark Awarded
1.	11	
2.	7	
3.	9	
4.	11	
5.	8	
6.	14	
7.	11	
8.	9	
Total	80	

ADDITIONAL MATERIALS

In addition to this paper you may require a calculator and a ruler.

INSTRUCTIONS TO CANDIDATES

Use black ink or black ball-point pen. Do not use gel pen. Do not use correction fluid.
Write your name, centre number and candidate number in the spaces at the top of this page.
Answer **all** questions.
Write your answers in the spaces provided in this booklet. If you run out of space, use the additional page at the back of the booklet, taking care to number the question(s) correctly.

INFORMATION FOR CANDIDATES

The number of marks is given in brackets at the end of each question or part-question.
Question 5(b) is a quality of extended response (QER) question where your writing skills will be assessed.



JUN193400U10101

Answer all questions.

1. (a) Complete the sentences below which describe how the bodies of animals and plants are built up and organised. [3]

Use words from the list below.

membranes organisms cells proteins organs

Similar are grouped together to form tissues.

Different types of tissues make up which are organised into systems.

The systems work together in such as animals and plants.

- (b) Animals and plants have specialised cells.

- (i) Identify the scientific term for the process which occurs in organisms resulting in cells becoming specialised. Choose your answer from the list. [1]

- A division
- B diffusion
- C differentiation
- D diversification

Answer ☐

- (ii) Explain how and why specialised cells are beneficial to organisms. [2]

.....

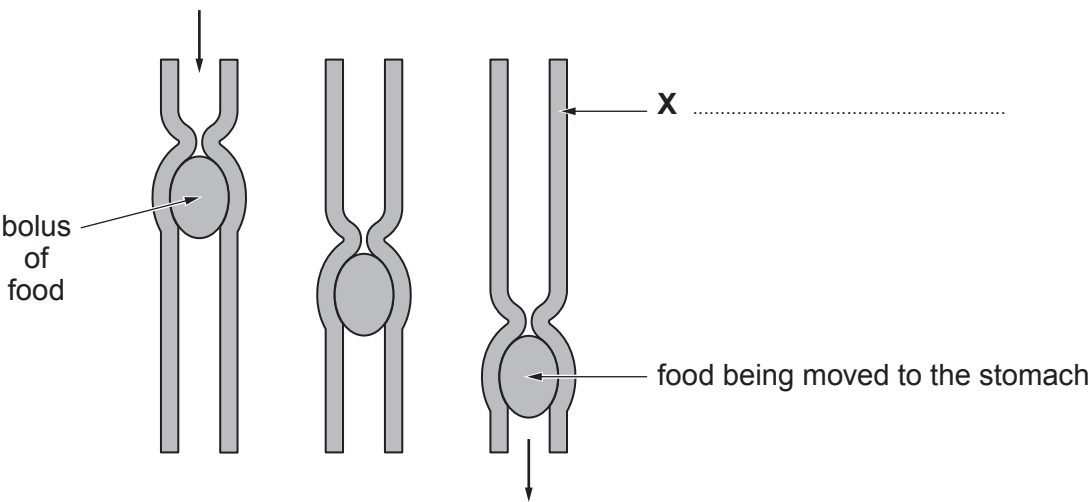
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- (c) The diagram shows the process of peristalsis in the human body.
Some food has just been swallowed.



- (i) Label structure **X** on the diagram. [1]
- (ii) State the name of the system in the human body to which structure **X** belongs. [1]
.....
- (iii) Name the specialised cells in the walls of structure **X** which enable peristalsis to occur. [3]
Describe how they cause the food to be moved during peristalsis.

.....

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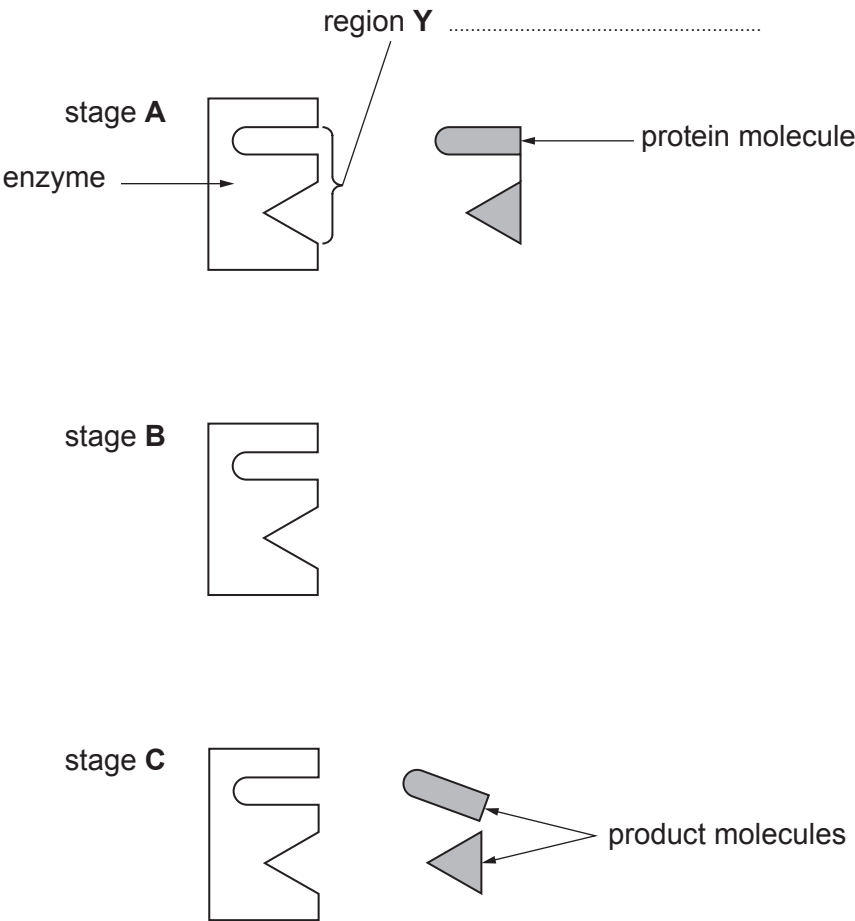
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2. The lock and key model is used to explain how enzymes work. It was first suggested in 1894 by Emile Fischer.

The diagrams below show stages in the action of an enzyme which digests protein molecules.



- (a) (i) **Label** region **Y** on the diagram. [1]
- (ii) **Complete the diagram** to show what occurs at stage **B**. [1]



Examiner
only

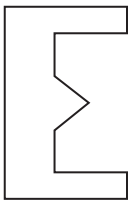
(b) (i) State the name of the **type** of enzyme shown in stage **A**. [1]

.....

(ii) State the name of the product molecules shown in stage **C**. [1]

.....

(c) The diagram below shows the same enzyme but it has been denatured.



(i) Explain why this denatured enzyme would not function. [2]

.....
.....
.....
.....

(ii) Suggest how the enzyme molecule shown could have become denatured. [1]

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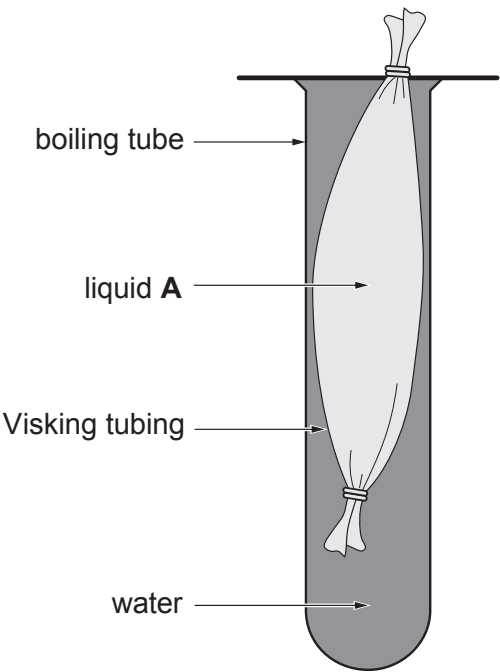
3400U101
05

7



3. The cell membrane controls which substances enter and leave the cell. Visking tubing can be used as a model of the cell membrane.

The diagram shows the apparatus used at the start of an investigation using Visking tubing.
The contents of liquid **A** were tested for glucose and starch at the start of the investigation.
Liquid **A** and the water in the boiling tube were tested for glucose and starch at 30 minutes.



(a) (i) Complete the table below to show the results of tests made on liquid **A** at the **start of the investigation**. [2]

Test	Colour change	Conclusion
Benedict's reagent to test for glucose	blue to	glucose present
Iodine solution to test for starch	brown to	starch present

(ii) Describe how you would test the water in the boiling tube for the presence of glucose using Benedict's reagent.

(You do not need to describe the result) [1]

.....

.....



(b) The table below shows the results for tests carried out on liquid **A** and the water in the boiling tube at 30 minutes.

Results at 30 minutes

	Test for glucose	Test for starch
liquid A	glucose present	starch present
water in boiling tube	glucose present	no starch present

Explain why, at 30 minutes, glucose molecules are present in the water in the boiling tube but starch molecules are not. [3]

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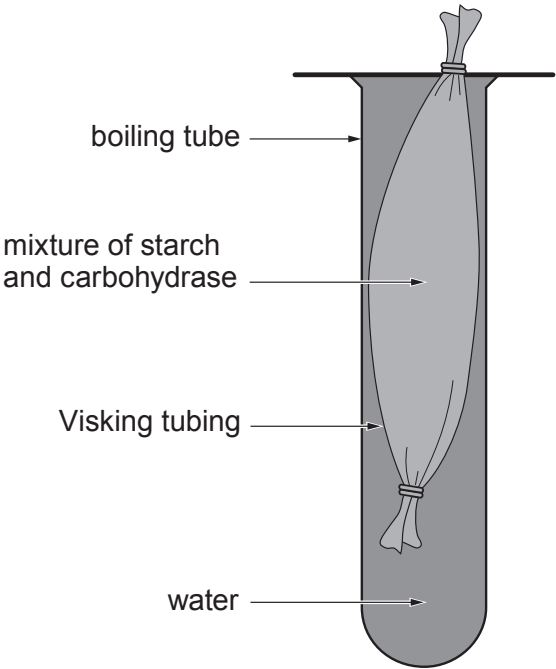
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(c) In a similar investigation, the Visking tubing contained **only** starch and the enzyme carbohydrase, as shown below.



The water in the boiling tube was tested at 30 minutes for the presence of glucose using Benedict's reagent.

Describe the result you would expect to observe. Explain your answer. [3]

.....

.....

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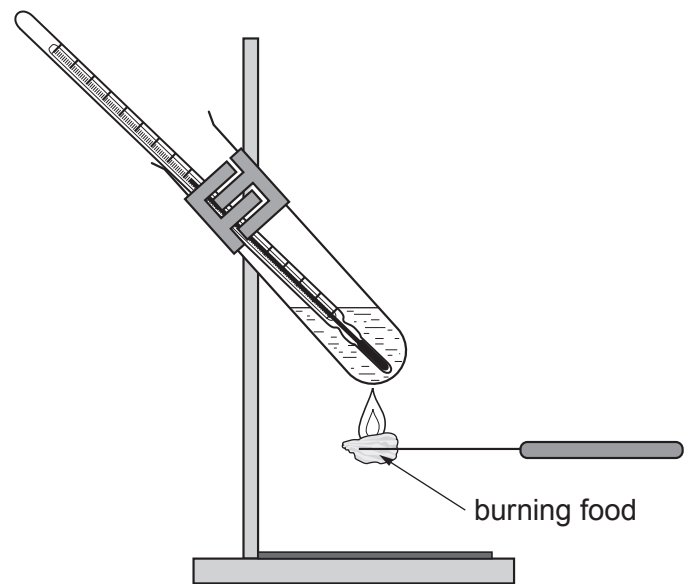
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4. (a) The apparatus shown in the diagram below can be used to compare the energy content of different foods.

The energy content of the food is released when it is burned and this causes the water to increase in temperature.



You are asked to use the apparatus to compare the energy content of different pasta types of known mass.

- (i) The table below shows part of the risk assessment for this investigation.

Complete the table. [2]

Hazard	Risk	Control measure
Apparatus is hot		

- (ii) For this investigation, state:

I. the apparatus you would use to find the mass of the pasta; [1]

.....



Examiner
only

II. **two** factors which you would keep constant throughout the investigation to ensure fair testing; [2]

.....

.....

III. how you would work out the increase in the temperature of the water. [2]

.....

.....

.....

.....

(iii) Explain why it is important to burn each piece of pasta completely. [1]

.....

.....

(b) The table shows some results from an investigation.

Pasta type	Total mass of pasta burned (g)	Total energy released from pasta (J)	Energy released in Joules per gram (J/g)
plain	9	293.4	32.6
wholewheat	10	282.0	28.2
green	9	252.0	

(i) Calculate the energy released in Joules per gram for green pasta. **Write your answer in the table.** [2]

Space for working.

(ii) From these results, state which type of pasta has the highest energy content. Give the reason for your answer. [1]

Type of pasta

Reason

11



Examiner
only

- 5. (a)** Complete the word equation for aerobic respiration in a human. [2]

Glucose + $\longrightarrow$ Carbon dioxide + + Energy

- (b) The table shows the gases present in inspired air (air breathed in) and expired air (air breathed out).

Gas	Inspired air (%)	Expired air (%)
oxygen	21.00	16.00
carbon dioxide	0.04	4.00
nitrogen	78.00	78.00
water vapour	varies	saturated

Compare the composition of inspired air with expired air **for each of the four gases** in the table. Explain why the differences occur. [6 QER]

3



6. Radish is a salad vegetable. The plants are grown in large quantities in greenhouses by commercial producers.

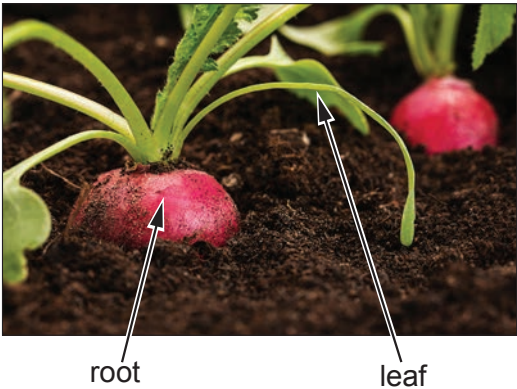
It is important that photosynthesis occurs at a high rate in the leaves of the plants to give the largest possible crop.

Photosynthesis requires carbon dioxide, water and light and produces glucose and oxygen.

Commercial greenhouse

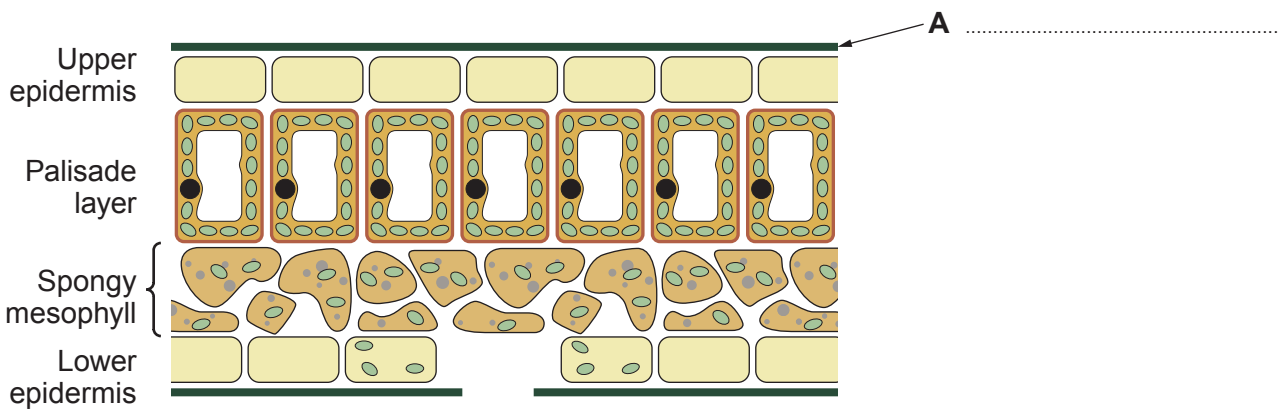


Radish plants



- (a) (i) The root of the radish contains stored starch. State the name of the molecules which are converted into starch. [1]

.....
The diagram below shows a section through a leaf of a plant such as radish.

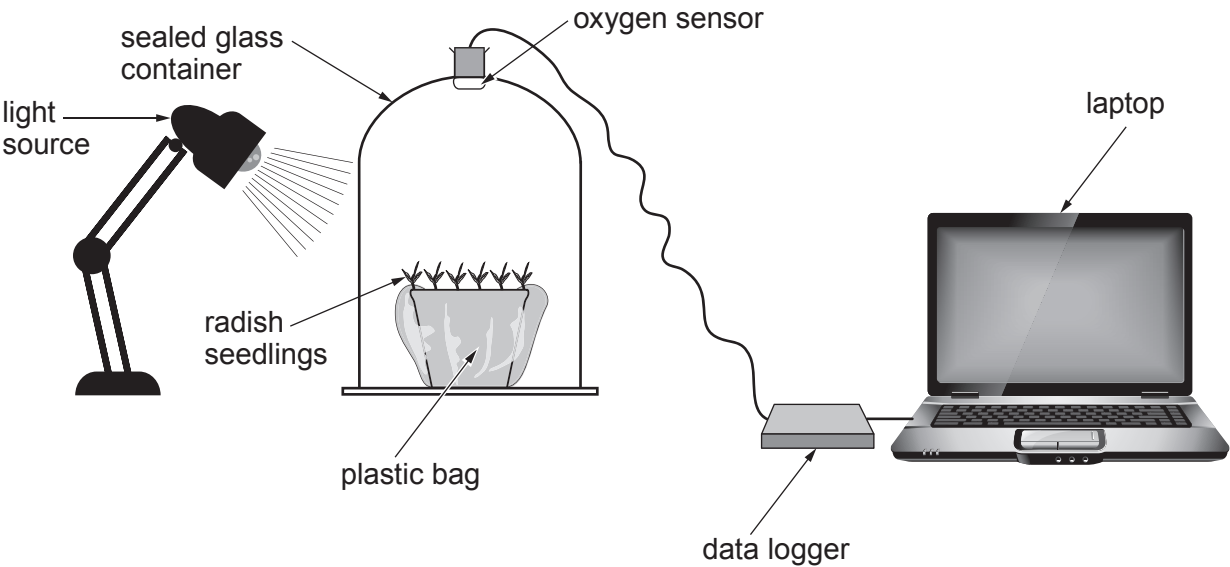


- (ii) I. **Label** structure **A** on the diagram. [1]
II. Explain why **most** photosynthesis occurs in the cells of the palisade layer. [3]

.....
.....
.....
.....
.....
.....



(b) Robert, a commercial radish grower, investigated the effects of temperature on the growth of radish seedlings. He used a range of temperatures and used the apparatus shown below to measure oxygen production. From this he calculated the mean rate of photosynthesis at each temperature.



The results of his investigation are shown on the graph and in the table.

Temperature (°C)	Rate of photosynthesis (a.u.)
5	<i>Values shown on the graph</i>
10	
20	
25	51
30	60
35	64
40	59
45	24
55	15

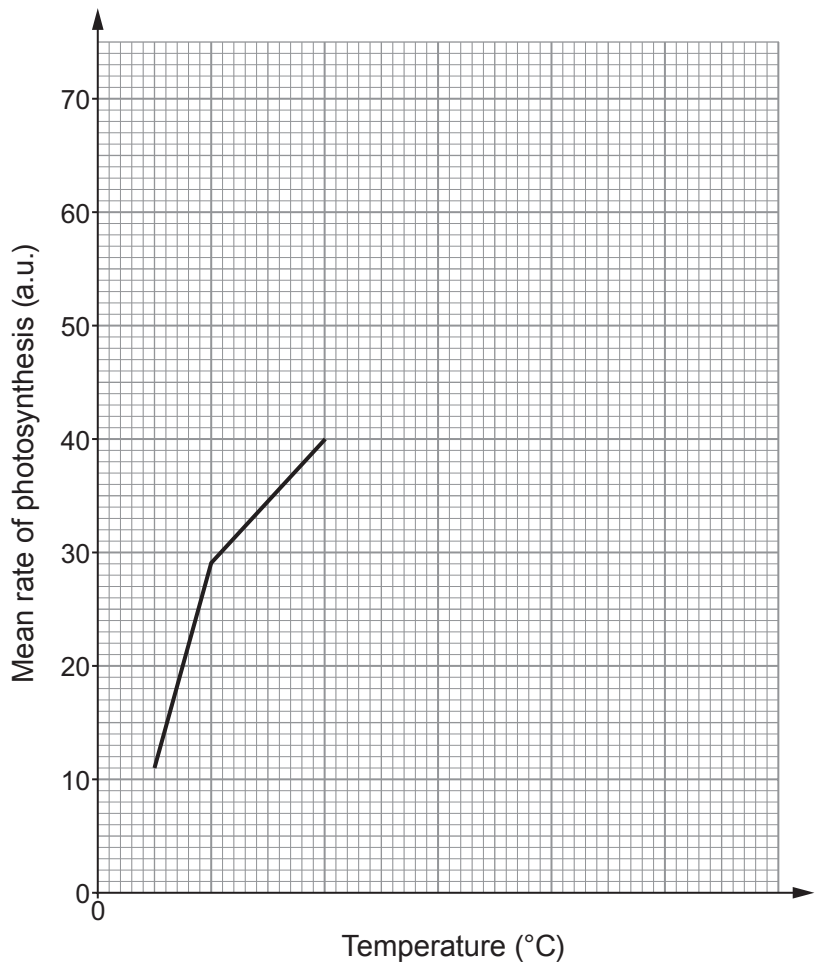
- (i) Complete the graph of results by: [4]
- I. adding the scale for temperature;

II. plotting the values from 25°C - 55°C;

III. joining the plots with a ruler to complete the line



Examiner
only



- (ii) From the graph,
- I. Describe how the increase in temperature affects the rate of photosynthesis. [2]
.....
.....
.....
.....
 - II. Calculate the difference in the rate of photosynthesis between 5°C and 20°C. [1]

Difference = a.u.



Examiner
only

(iii) I. From these results state the optimum temperature for Robert to use for growing radish plants and explain your answer. [1]

.....

.....

II. How could Robert improve his investigation so that the optimum temperature could be identified more accurately? [1]

.....

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14



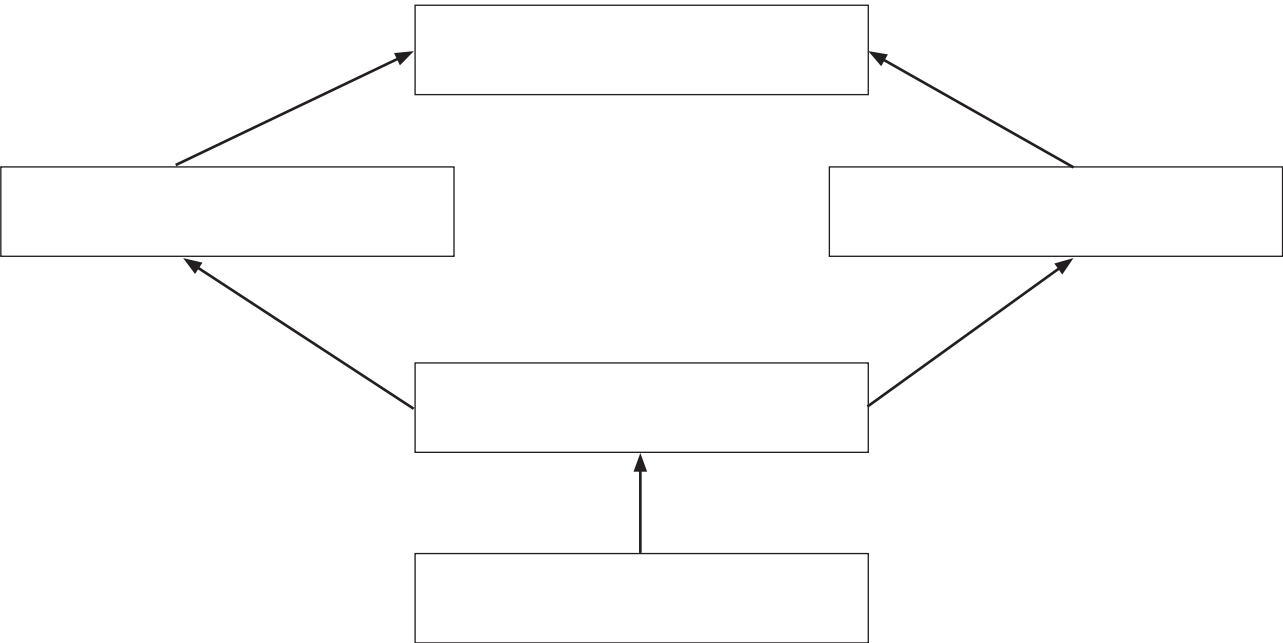
7. Killer whales (*Orcinus orca*) feed on thousands of large fish and squid. These large fish and squid feed on hundreds of small fish. Small fish feed on tiny organisms called plankton.

In 2016, a dead killer whale was found washed up on an island off the West Coast of Scotland. Tests revealed the whale contained one of the highest levels of PCBs (polychlorinated biphenyls) ever recorded. Levels of PCBs are measured in milligrams per kilogram of fat in an animal's body. Scientists believe that the threshold where PCBs can cause harm to killer whales is 30 mg/kg of fat. The level of PCB in the dead whale was 957 mg/kg of fat.

During the last century, PCBs were widely used in plastics, electrical goods and cement. PCBs were banned in the 1970s. It is estimated that there are a million tonnes of PCB-contaminated material waiting to be disposed of in Europe. PCBs are released into the environment from decomposing products that have not been disposed of properly. Plankton absorb PCBs that are washed into the oceans.

There are only eight resident killer whales remaining in UK waters. Scientists have not observed any offspring being born to the resident killer whales in the 25 years they have been studying them.

(a) (i) Complete the simple food web below to show the feeding relationships of a killer whale. [3]



(ii) State the term used to describe the feeding stage of killer whales. [1]

.....



Examiner
only

- (b) Calculate how many times greater the level of PCBs was in the dead killer whale’s body when compared to the threshold where PCBs cause harm. [1]

Times greater =

- (c) Suggest the name of a group of micro-organisms responsible for releasing PCBs into the environment. [1]

.....

- (d) Explain how the PCBs led to the death of the killer whale. [2]

.....
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- (e) Apart from causing death, describe another effect of PCBs on killer whales in UK waters. Give evidence to support your answer. [2]

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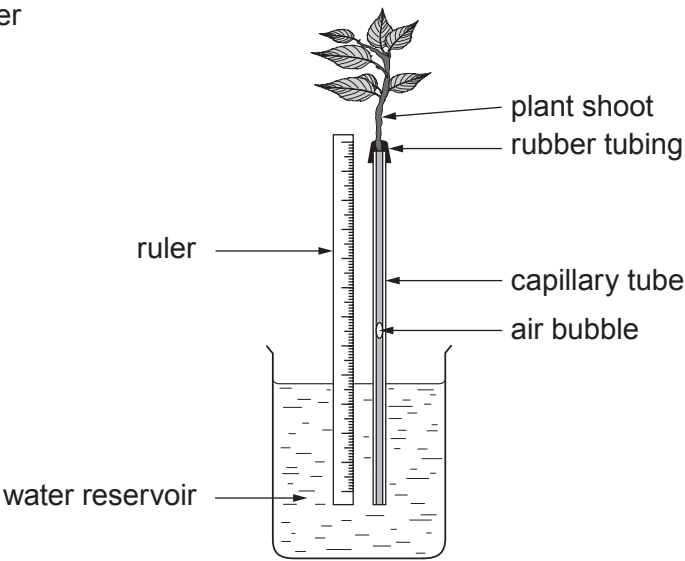
- (f) Suggest why PCBs are still present in UK waters even though they have been banned since the 1970s. [1]

.....
.....



8. Students used a simple potometer as shown below to compare the rate of transpiration in shoots from a beech tree (*Fagus sylvatica*) and a bay tree (*Laurus nobilis*).

A simple potometer



(a) State the name of the tissue that transports water in plants. [1]

.....

(b) State **two** uses of water in plants. [2]

.....
.....

(c) (i) When the beech tree shoot was used, the air bubble in the capillary tube took 87 seconds to travel 50 mm. Calculate the rate of transpiration for the shoot from the beech tree in mm/s. [1]

Rate of transpiration = mm/s

(ii) When the bay tree shoot was used, the air bubble in the capillary tube took 51 minutes and 58 seconds to travel 50 mm.

State **one** conclusion that could be reached from the results. [1]

.....
.....



Examiner
only

(d) (i) State **one** factor that should have been kept constant in this investigation. [1]

.....

(ii) Describe how the students could improve their confidence in the results. [1]

.....

.....

(e) Explain why a potometer cannot be used to study transport of sugars in a plant. [2]

.....

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9

END OF PAPER



Surname	Centre Number	Candidate Number
First name(s)		0

**GCSE**

3400U10-1



Z22-3400U10-1

WEDNESDAY, 15 JUNE 2022 – MORNING

BIOLOGY – Unit 1:
Cells, Organ Systems and Ecosystems

FOUNDATION TIER

1 hour 45 minutes

For Examiner's use only		
Question	Maximum Mark	Mark Awarded
1.	5	
2.	6	
3.	7	
4.	10	
5.	12	
6.	9	
7.	11	
8.	9	
9.	11	
Total	80	

3400U101
01**ADDITIONAL MATERIALS**

In addition to this paper you may require a calculator and a ruler.

INSTRUCTIONS TO CANDIDATES

Use black ink or black ball-point pen. Do not use gel pen or correction fluid.

You may use pencil for graphs and diagrams only.

Write your name, centre number and candidate number in the spaces at the top of this page.

Answer **all** questions.

Write your answers in the spaces provided in this booklet. If you run out of space, use the additional page at the back of the booklet, taking care to number the question(s) correctly.

INFORMATION FOR CANDIDATES

The number of marks is given in brackets at the end of each question or part-question.

Question 7(a) is a quality of extended response (QER) question where your writing skills will be assessed.

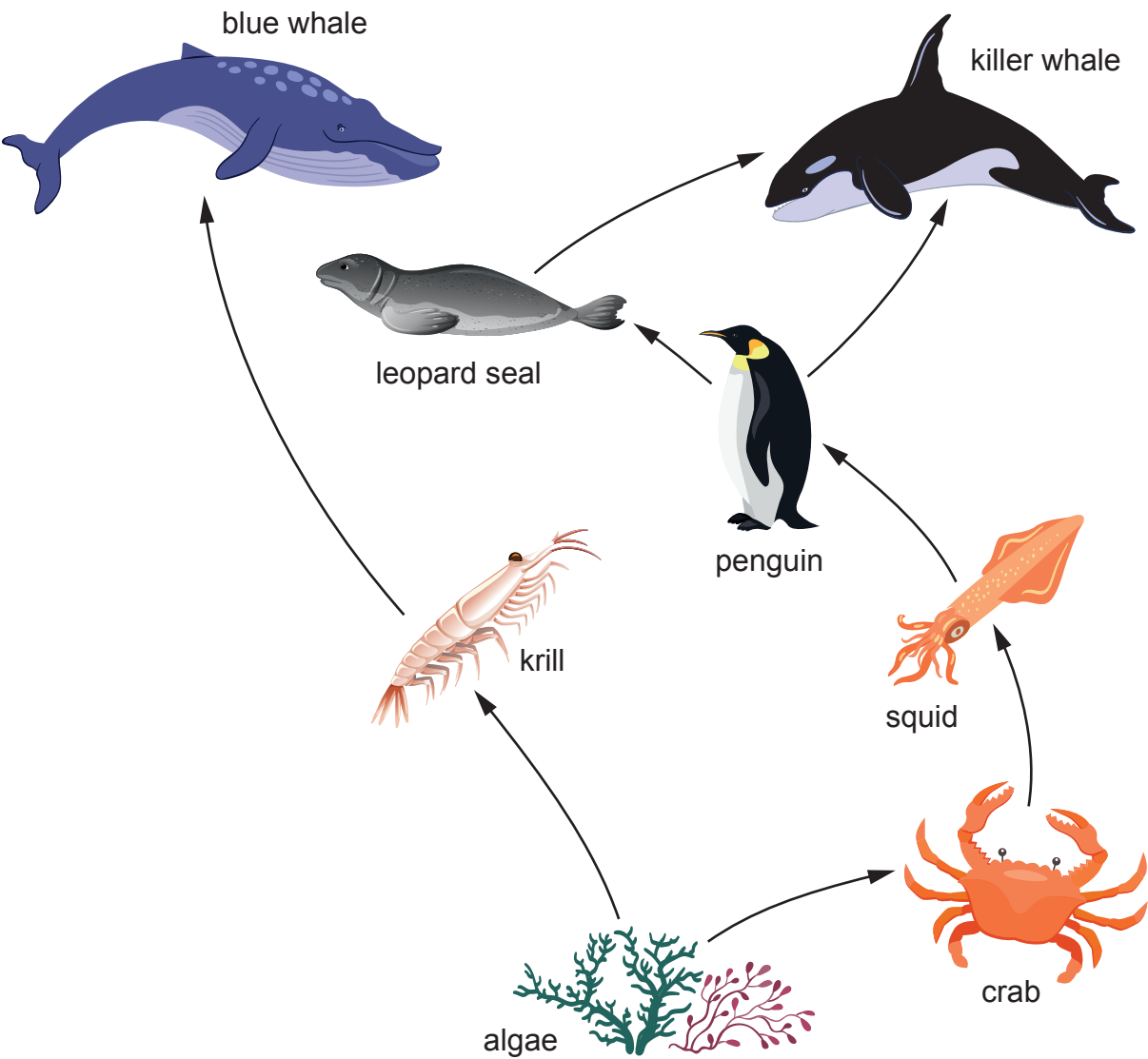


JUN223400U10101

Answer **all** questions.

1. **Image 1.1** shows a marine food web.

Image 1.1



Images are not to scale.



Examiner
only

- (a) From the food web in **Image 1.1**, state the name of:
- (i) the producer; [1]
.....
- (ii) the tertiary (third-stage) consumer. [1]
.....
- (b) State what the arrows in the food web in **Image 1.1** represent. [1]
.....
- (c) A disease caused the number of squid to decrease. Suggest what would happen to the numbers of penguins and crabs by underlining the correct word in each of the following sentences.
- (i) The number of penguins would (**increase** / **decrease**). [1]
- (ii) The number of crabs would (**increase** / **decrease**). [1]

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03

5



2. Images 2.1A, 2.1B and 2.1C show a root hair cell, xylem and phloem, and a leaf.

Image 2.1A

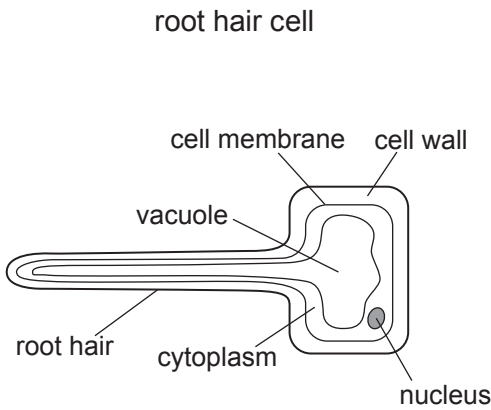


Image 2.1B

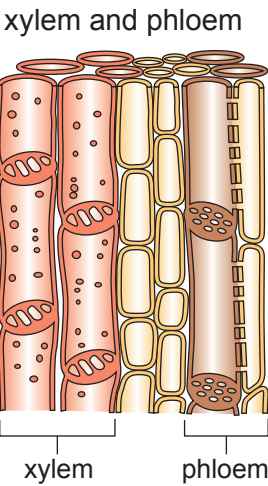


Image 2.1C



Images are not to scale.

(a) **Complete Table 2.2** by matching the parts of the plant to their functions, **A–D** shown below. [3]

- A** – transports sucrose around the plant
- B** – absorbs water from the soil
- C** – transports water in the plant
- D** – absorbs light for photosynthesis

Table 2.2

Part of plant	Function (A–D)
root hair cell	
xylem	
phloem	
leaf	



(b) Complete the following sentences by choosing some of the words below. [3]

similar cells different organ organism specialised

A root hair cell is an example of a cell. Xylem and phloem are both tissues. Tissues are groups of similar with a similar function. A leaf is an which is made up from different tissues working together.

6

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05

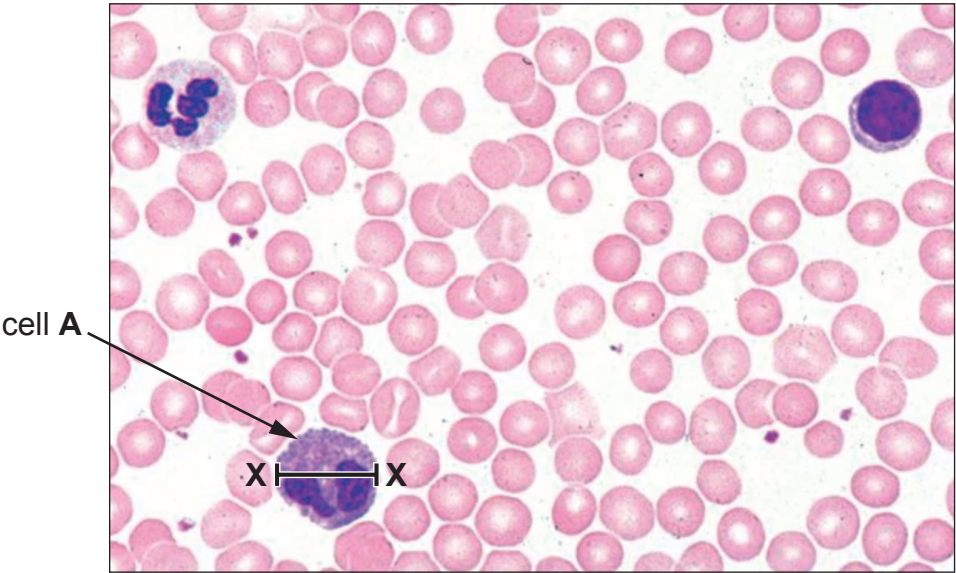


3. (a) Use a ruler to draw lines to match up the parts of blood with their functions. [2]

Part of blood	Function
plasma ●	● carry oxygen
white blood cells ●	● defend against disease
red blood cells ●	● clot blood
platelets ●	● carries carbon dioxide and dissolved food

(b) Image 3.1 shows a sample of human blood as seen down a microscope.

Image 3.1



Examiner
only

(i) Measure the length of the line **X–X** in **Image 3.1**. [1]

Length of line **X–X** = mm

(ii) The magnification of **Image 3.1** is $\times 1\,000$. Use the following equation to calculate the actual size of cell **A**. [2]

Actual size of cell **A** = $\frac{\text{measured length of line X–X}}{\text{magnification of Image 3.1}}$

Space for working

Actual size of cell **A** = mm

(iii) Identify the type of blood cell labelled **A** and give a reason for your answer. [2]

Type of blood cell

Reason

3400U101
07

7



4. A scientist was investigating changes in lactic acid concentration in the blood of two students, Cerys and Blodwen. They both ran on a treadmill at the same speed for the same time.

(a) **Table 4.1** shows the results for Cerys.

Table 4.1

Time (minutes)	Concentration of lactic acid (mg per dm ³)
0	60
10	120
20	180
30	200
40	150
50	110
60	70
70	60

(i) Complete **Graph 4.2** of these results by:

- Plotting the concentration of lactic acid for Cerys [2]
- Joining your plots with a ruler. [1]

The first three plots for Cerys and all the plots for Blodwen have been done for you.

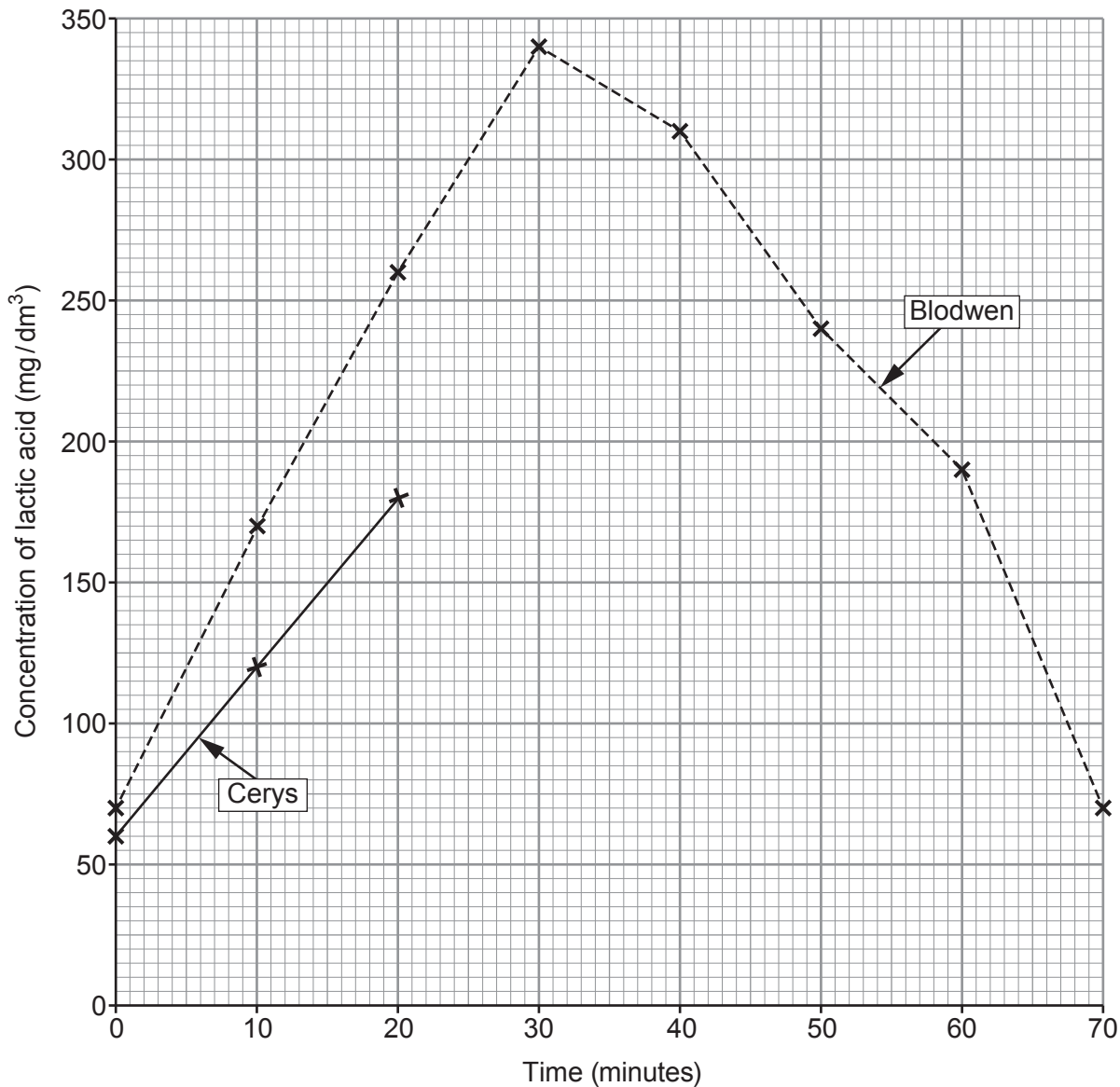
(ii) From **Graph 4.2**, describe **one** difference between the results for Cerys and Blodwen. [1]

.....

.....



Graph 4.2



(b) Use **Graph 4.2** to:

(i) calculate the maximum increase in lactic acid in the blood for Blodwen; [1]

Increase = mg per dm³

(ii) suggest what time Blodwen stopped running. [1]

Time =minutes

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- (c) The students are using anaerobic respiration for part of the time they are running. Complete the word equation for anaerobic respiration. [2]

..... → (+ energy)

- (d) The students continued to breathe deeply and rapidly after they had finished running. Explain the reason for this. [2]

.....

.....

.....

.....

10

5. The amount of pollution in water can be determined by measuring oxygen levels, pH levels or by using indicator species. Polluted water will have low oxygen levels, as bacteria use up oxygen. The water will become more acidic.

Air pollution can be assessed using indicator species, such as lichens. Lichens are organisms that consist of fungi and algae living together. Lichens can grow on tree bark and rocks. There are many species of lichen which can be grouped into three main types: bushy, leafy and crusty. The number of types of lichens present can indicate the level of air pollution.

- Bushy lichens can only survive in unpolluted air.
- Leafy lichens can survive a small amount of air pollution.
- Crusty lichens can survive in more polluted air.
- The more types of lichen growing in an area, the lower the level of air pollution.

- (a) Use the information above to **complete Table 5.1** by writing true or false next to each statement. [4]

Table 5.1

Statement	True/False
Bacteria use oxygen for respiration	
Polluted water will be pH 7 (neutral)	
Lichens are plants	
There will be a higher number of different types of lichen in polluted areas	
Bushy lichens can survive in more polluted areas	



- (b) (i) In 2019, a study rated the air quality in the centre of Cardiff as being worse than the centre of London.

Image 5.2 shows the three main types of lichen.

In **Image 5.2** circle the letter (**A, B, C**) of the type of lichen you may expect to find growing in Cardiff city centre if there was a lot of air pollution present. [1]

Image 5.2

A



Bushy

B



Leafy

C



Crusty

A scientist counted the number of species of lichens growing at different distances from the centre of Cardiff. Her results are shown in **Table 5.3**.

Table 5.3

Distance from Cardiff centre (km)	Number of species of lichen
0	2
1	5
2	8
3	10
4	15
5	
6	26
7	30
8	48
9	56
10	60

- (ii) **Complete Table 5.3** by estimating the number of species of lichen likely to be found 5 km from Cardiff centre. [1]
- (iii) State the independent and dependent variables in this investigation. [2]

Independent variable

Dependent variable



Examiner
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(iv) I. State the relationship between the number of lichen species and the distance from the centre of Cardiff. [1]

.....

.....

II. Use the information given to explain the relationship between the number of species of lichen and the distance from the centre of Cardiff. [2]

.....

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.....

(v) State how the scientist could improve her confidence in these results. [1]

.....

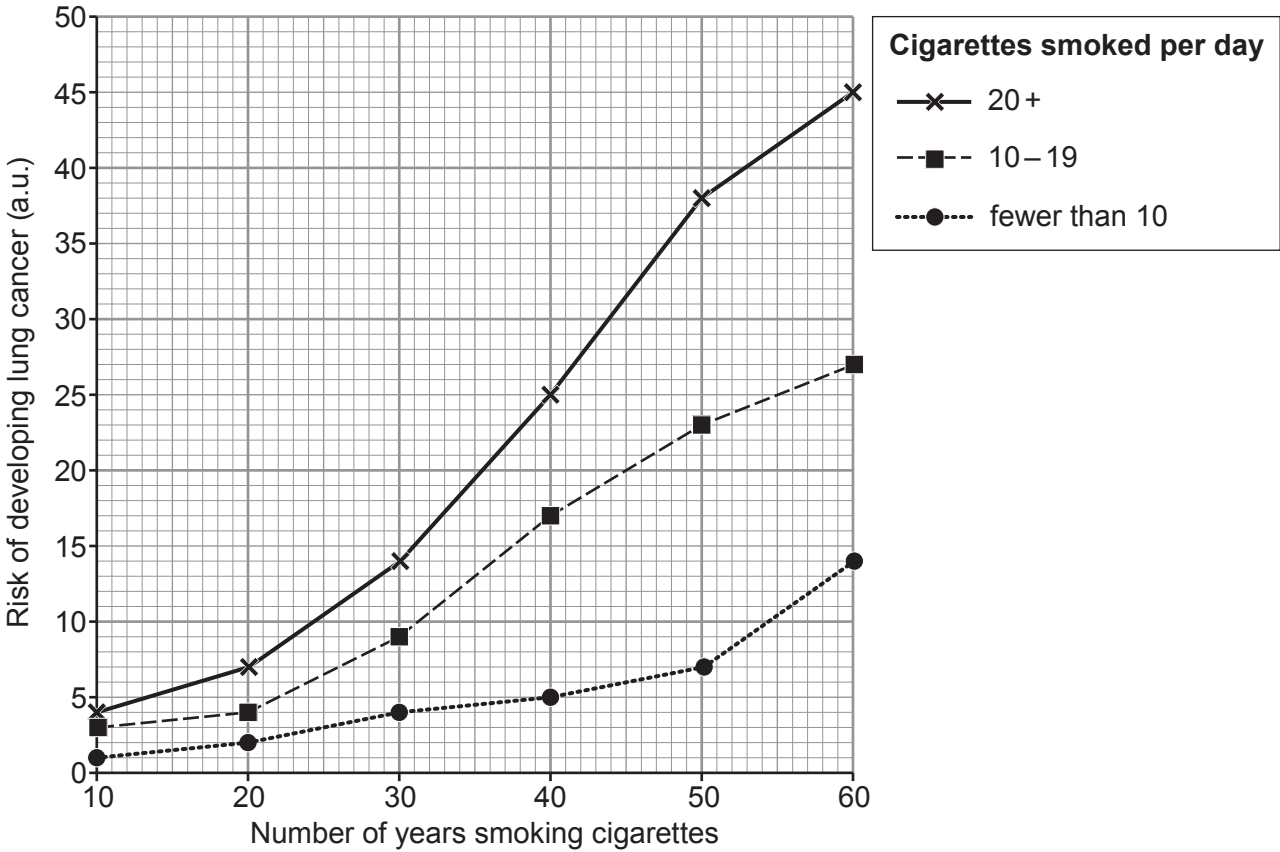
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6. **Graph 6.1** shows the relationship between smoking cigarettes and the risk of developing lung cancer.

Graph 6.1



(a) State **two** conclusions that can be drawn from **Graph 6.1**. [2]

1.

.....

2.

.....

(b) State **one** chemical found in cigarette smoke and its effect on the body. [2]

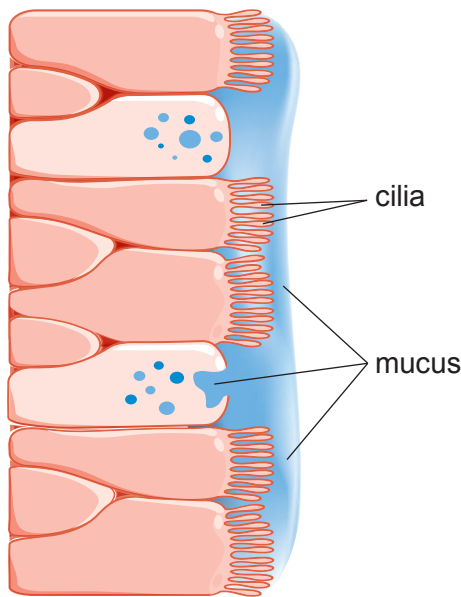
Chemical

Effect



(c) **Image 6.2** shows the cells that line the airways of the lungs.

Image 6.2



(i) Describe the function of:

I. mucus; [1]

.....

II. cilia. [1]

.....

(ii) Describe the effect of smoking on the cilia and mucus. [2]

.....

.....

.....

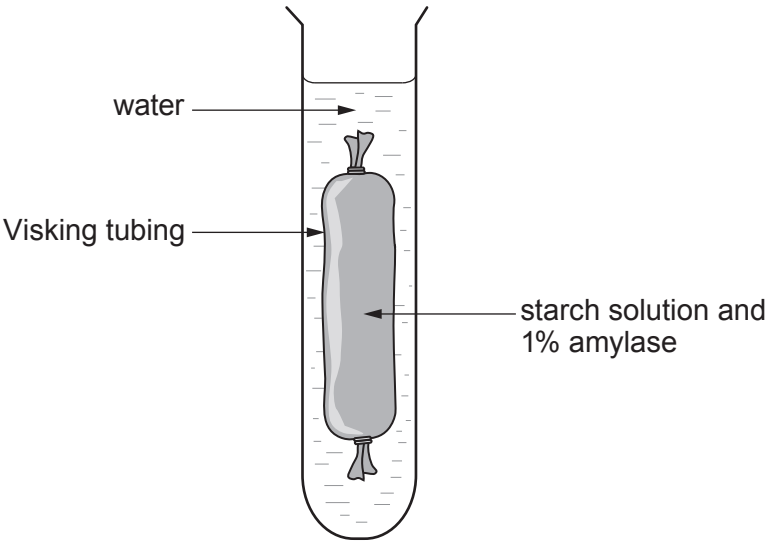
(iii) Other than lung cancer, state the name of **one** disease which can be caused by cigarette smoking. [1]

.....



7. An experiment was set up using Visking tubing as a model gut. The Visking tubing was filled with a solution containing starch and the enzyme amylase.

Image 7.1



- (a) The water surrounding the Visking tubing was tested for starch and glucose at the start and at 30 minutes. The results are shown in **Table 7.2**.

Table 7.2

Substance	At start	At 30 minutes
glucose	absent	present
starch	absent	absent



Describe the chemical tests for starch and glucose. Explain the results obtained in this experiment at **30 minutes**. [6 QER]

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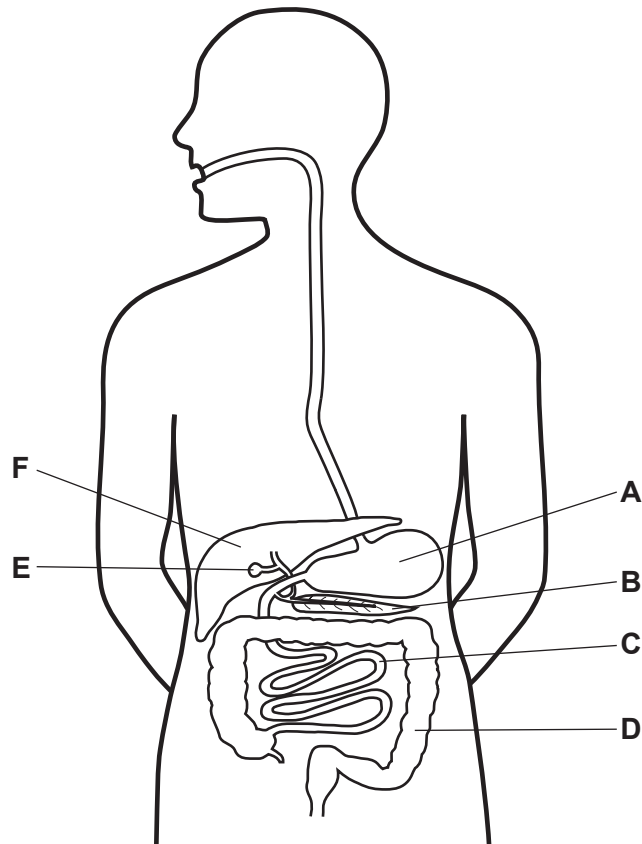
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(b) **Image 7.3** shows the human digestive system.

Image 7.3



- (i) State the letter of the organ (**A–F**) where bile is: [2]
- I. made;
- II. stored.

- (ii) Explain the function of bile in the digestion of fat. [2]

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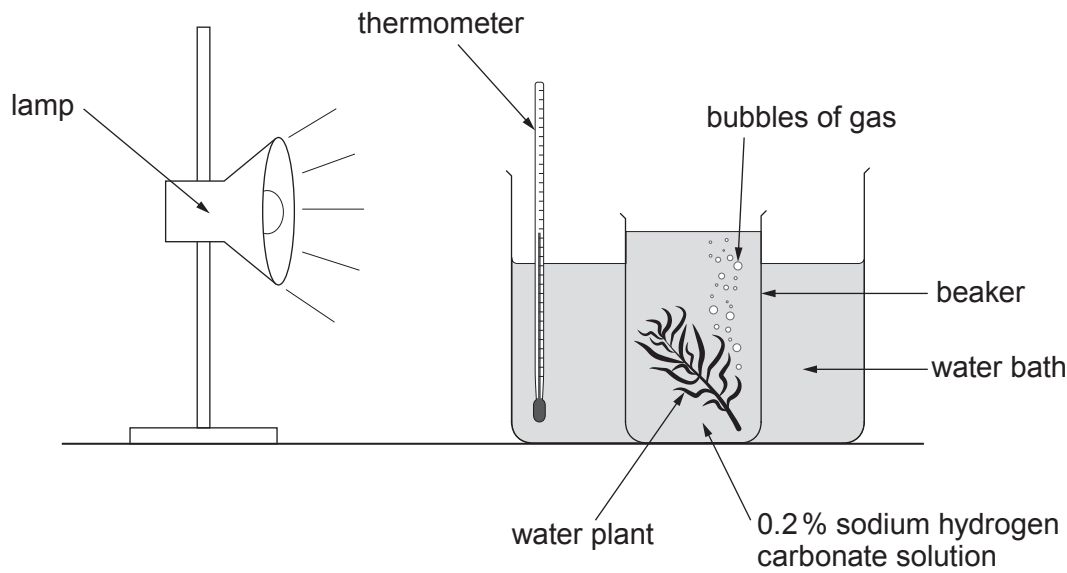
- (iii) State the name of the organ where digestion of fat takes place. [1]

.....



8. Students were instructed to investigate the effect of temperature on photosynthesis. They used the apparatus shown in **Image 8.1**.

Image 8.1



- The number of bubbles of gas produced by the water plant per minute was counted at different temperatures.
- The experiment was carried out three times at each temperature.
- The results are shown in **Table 8.2**. Means were calculated to the nearest whole number.

Table 8.2

Temperature of water bath (°C)	Number of bubbles of gas produced per minute			
	Test 1	Test 2	Test 3	Mean
20	18	21	12	17
25	15	24	21	20
30	24	27	30	27
35	26	25	26	26
40	14	13	10	12
45	3	5	6	
50	0	0	0	0



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only

(a) (i) State the name of the gas produced during photosynthesis. [1]

.....

(ii) State **two** ways in which plants use the glucose produced during photosynthesis. [2]

I.

II.

(b) (i) Calculate the mean number of bubbles per minute for 45 °C. **Write your answer in Table 8.2.** [2]

(ii) Describe the relationship between the temperature and the number of bubbles produced per minute. [2]

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(iii) Explain the result for 50 °C. [2]

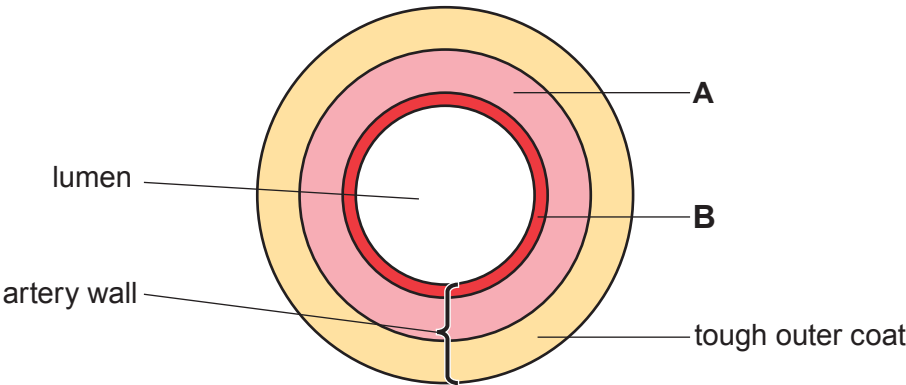
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9



9. Image 9.1 shows a cross section through a healthy coronary artery.

Image 9.1



(a) (i) State the names of **tissues A** and **B**. [2]

A

B

(ii) Suggest how the diameter of the lumen would differ in an individual with cardiovascular disease and explain the difference. [2]

.....

.....

.....

.....

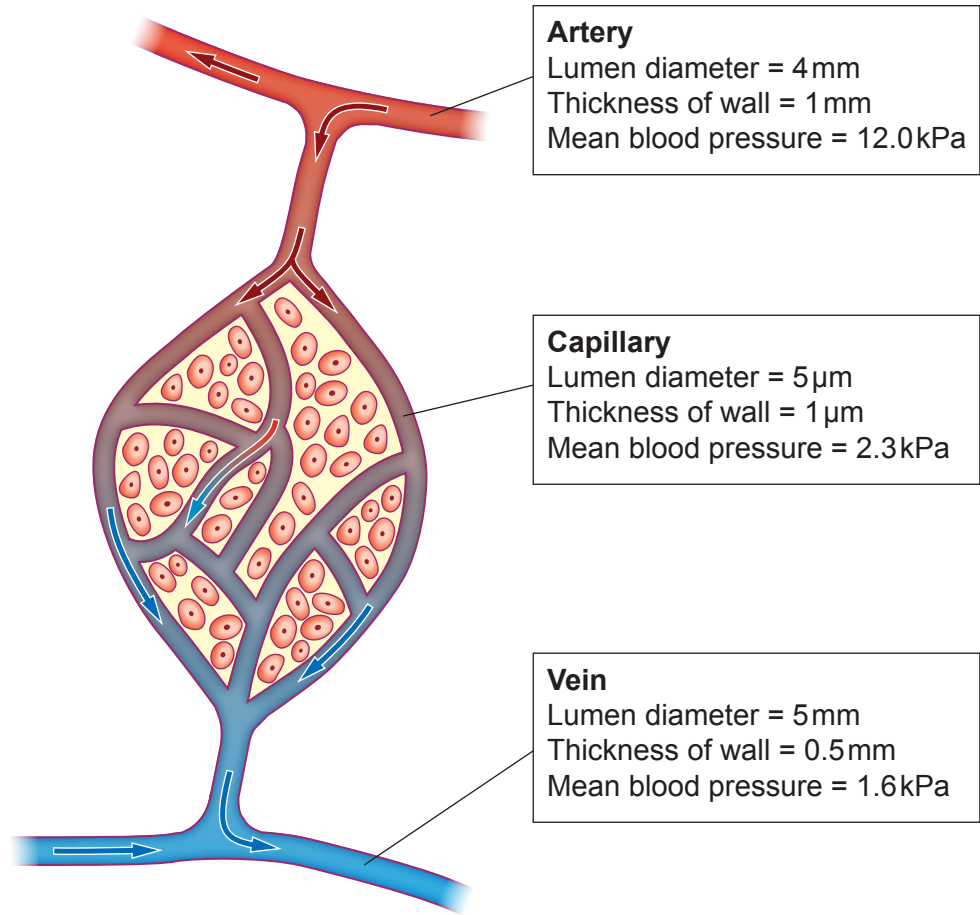
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- (b) **Image 9.2** shows a cross section of blood vessels found in muscle tissue, along with some information about them.

Image 9.2



- (i) Use **Image 9.2** to give **two** differences between the structure of an artery and a vein. [2]

.....

.....

.....

.....

- (ii) Veins contain structures that are not present in arteries or capillaries. State the name of these structures and describe their function. [2]

Name of structures

Function

.....



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(c) (i) State the name of **one** substance which passes from the blood into the body cells. [1]

.....

(ii) State **one** way in which capillaries are adapted for their function. Give an advantage for this adaptation. [2]

Adaptation

.....

Advantage

.....

.....

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END OF PAPER



Surname	Centre Number	Candidate Number
First name(s)		0



GCSE
3400U10-1



TUESDAY, 13 JUNE 2023 – MORNING

BIOLOGY – Unit 1:
Cells, Organ Systems and Ecosystems
FOUNDATION TIER
1 hour 45 minutes

For Examiner's use only		
Question	Maximum Mark	Mark Awarded
1.	8	
2.	8	
3.	10	
4.	12	
5.	7	
6.	9	
7.	6	
8.	12	
9.	8	
Total	80	

ADDITIONAL MATERIALS

In addition to this paper you may require a calculator and a ruler.

INSTRUCTIONS TO CANDIDATES

Use black ink or black ball-point pen. Do not use gel pen or correction fluid.
You may use a pencil for graphs and diagrams only.
Write your name, centre number and candidate number in the spaces at the top of this page.
Answer **all** questions.
Write your answers in the spaces provided in this booklet. If you run out of space, use the additional page at the back of the booklet, taking care to number the question(s) correctly.

INFORMATION FOR CANDIDATES

The number of marks is given in brackets at the end of each question or part-question.
Question 7 is a quality of extended response (QER) question where your writing skills will be assessed.

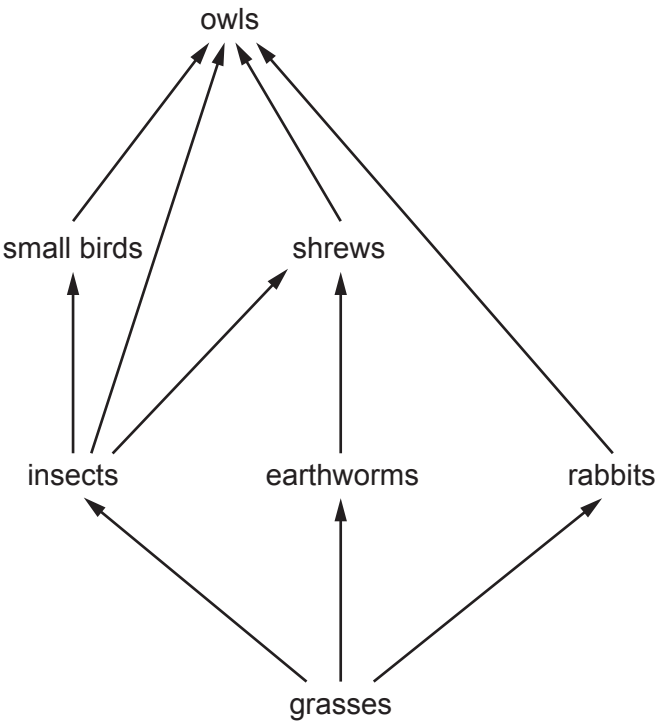


JUN233400U10101

Answer **all** questions.

1. (a) **Image 1.1** shows a food web from an area of grassland.

Image 1.1



(i) Use **Image 1.1** to state the organisms which are present in the highest numbers. [1]

(ii) Name the **third stage** (tertiary) consumers shown in the food web in **Image 1.1**. Give the reason for your choice. [2]

Name

Reason



Examiner
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(b) One food chain from the food web is shown below. The images are not to scale.

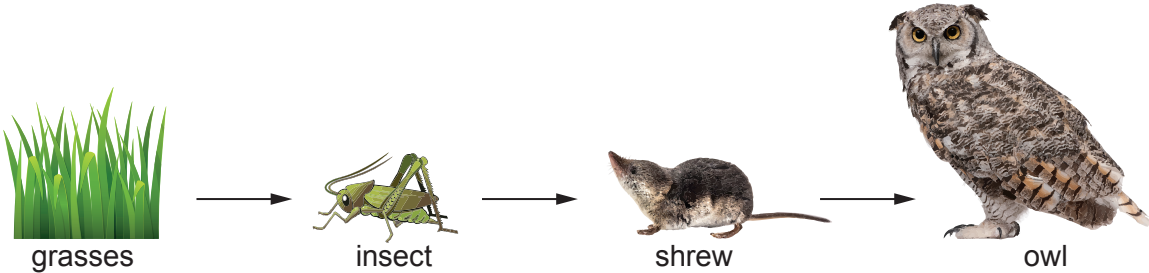


Table 1.2 shows the results of an investigation of the organisms in this food chain in the area of grassland habitat.

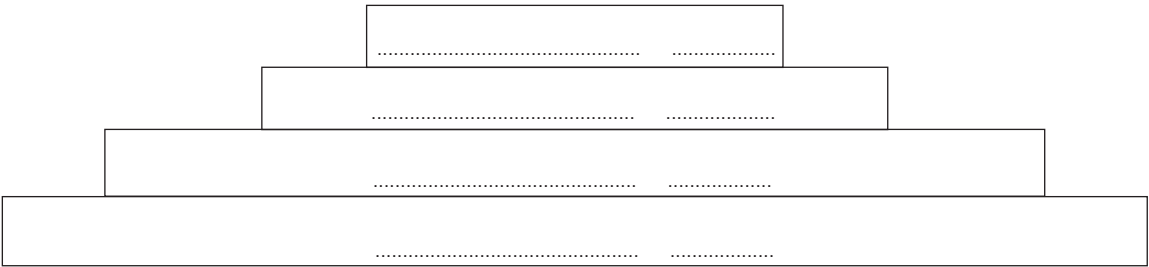
Table 1.2

Organism	Number of individuals counted	Mass of one individual (g)	Total biomass (g)
shrew	90	10	
owl	1	350	
grasses	10 000	2	20 000
insect	3 000	4	

(i) **Complete Table 1.2** by calculating the total biomass for each of the organisms in this food chain. One has been done for you. [2]
Space for working.

(ii) Use **Table 1.2** to complete the pyramid of numbers in **Image 1.3** by **writing the names of the organisms** in this food chain and the **numbers of each**. [1]

Image 1.3



(iii) State **one** way in which energy can be lost from a food chain and explain how this loss of energy is shown in the pyramid. [2]

.....

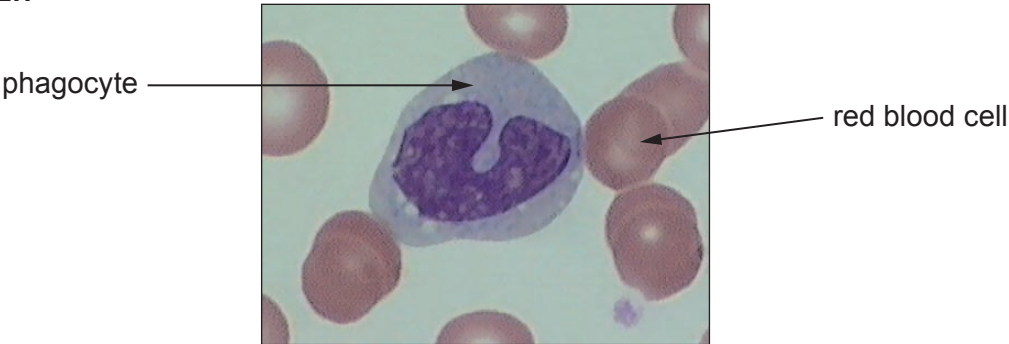
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2. **Image 2.1** shows a sample of human blood as seen through a light microscope.

Image 2.1

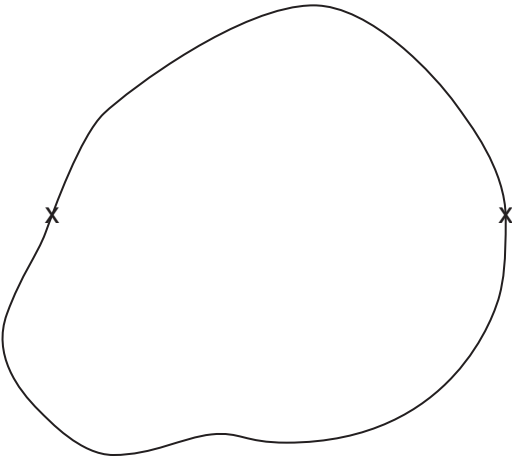


(a) (i) The drawing in **Image 2.2** shows the cell membrane of the phagocyte.

I. **Draw** the **nucleus** in the outline of the phagocyte and **label** it. [1]

II. **Label** the **cytoplasm** in the outline of the phagocyte. [1]

Image 2.2



(ii) I. Measure the width x-x **in mm** on the drawing. [1]

Width x-x = mm

II. The width of the actual cell is 0.012 mm.
Calculate the magnification of the drawing using the equation below. [1]

magnification = $\frac{\text{width x-x on drawing}}{\text{width of the actual cell}}$

magnification = ×



(b) **Complete the table** by selecting functions from those below.

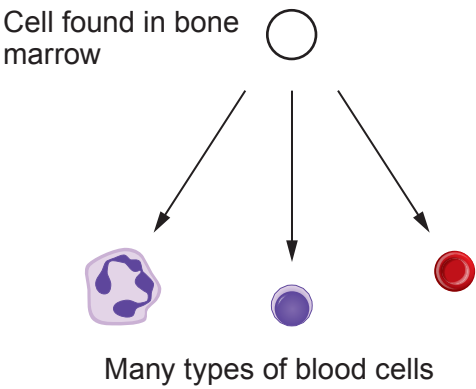
[3]

- transports hormones
- carries oxygen
- defends against disease
- clots the blood
- controls temperature

Part of blood	Function
red blood cell	
white blood cell	
platelets	

(c) There are many types of blood cell, each with a particular function. They all come from one type of cell found in bone marrow through the process of differentiation. This is shown in **Image 2.3**.

Image 2.3



Choose the letter **A-D** to give the term used for the various types of blood cells.

[1]

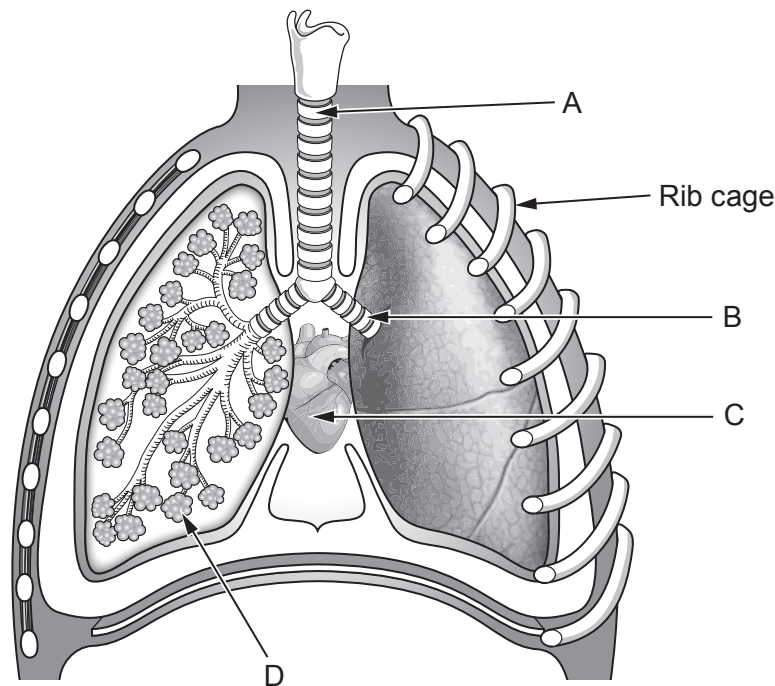
- A selected
- B specialised
- C secondary
- D sensitive

Answer letter =



3. (a) **Image 3.1** is a vertical section through the human chest cavity (thorax).

Image 3.1



Choose the letter (**A–D**) from **Image 3.1** which shows

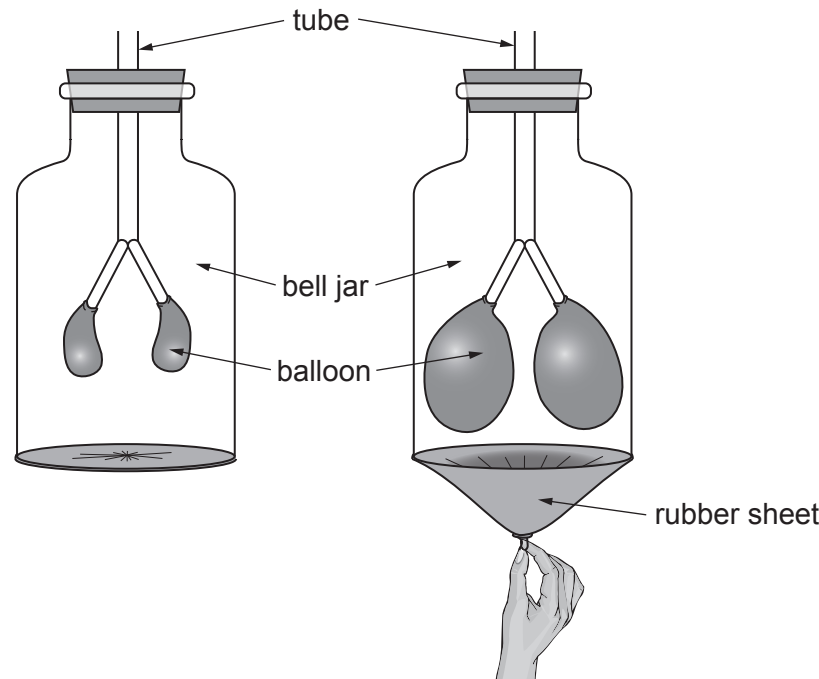
[2]

- (i) a bronchus
- (ii) an alveolus



- (b) The way in which the respiratory system works during breathing can be demonstrated using the bell-jar model shown in **Image 3.2**.

Image 3.2



- (i) State the part of the model shown in **Image 3.2** which represents: [3]
- I. the diaphragm
 - II. the lungs
 - III. the trachea
- (ii) The rubber sheet can be pulled downwards to demonstrate **inspiration** (breathing in).

Complete each of the sentences below by circling the correct statement to describe the process of **inspiration** in the human thorax. [2]

The volume of the thorax **increases** / **decreases** / **stays the same**.

The pressure in the thorax **increases** / **decreases** / **stays the same**.



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(iii) Describe how the ribcage moves during **inspiration** and why this cannot be demonstrated using the bell jar model. [2]

.....

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(c) Human muscle cells usually carry out aerobic respiration. During strenuous exercise, however, anaerobic respiration occurs.

State the harmful chemical substance produced by anaerobic respiration in human muscle cells. [1]

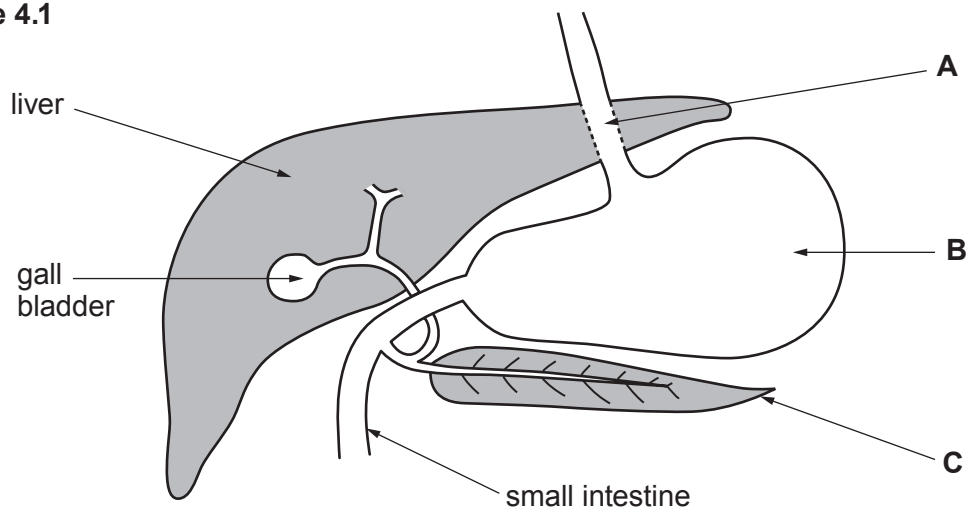
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4. **Image 4.1** shows part of the digestive system in a human.

Image 4.1



(a) (i) State the letter (**A-C**) which shows the pancreas. [1]

.....

(ii) The pancreas produces lipase.

Complete the sentence below using one of the terms given.

glucose glycerol amino acids

Lipase breaks down fat into fatty acids and [1]

(iii) The breakdown of fat by lipase is assisted by bile.
Using **Image 4.1** describe the pathway taken by bile as it passes from the **liver**,
where it is produced, to where it enters the **small intestine**. [2]

.....
.....
.....
.....



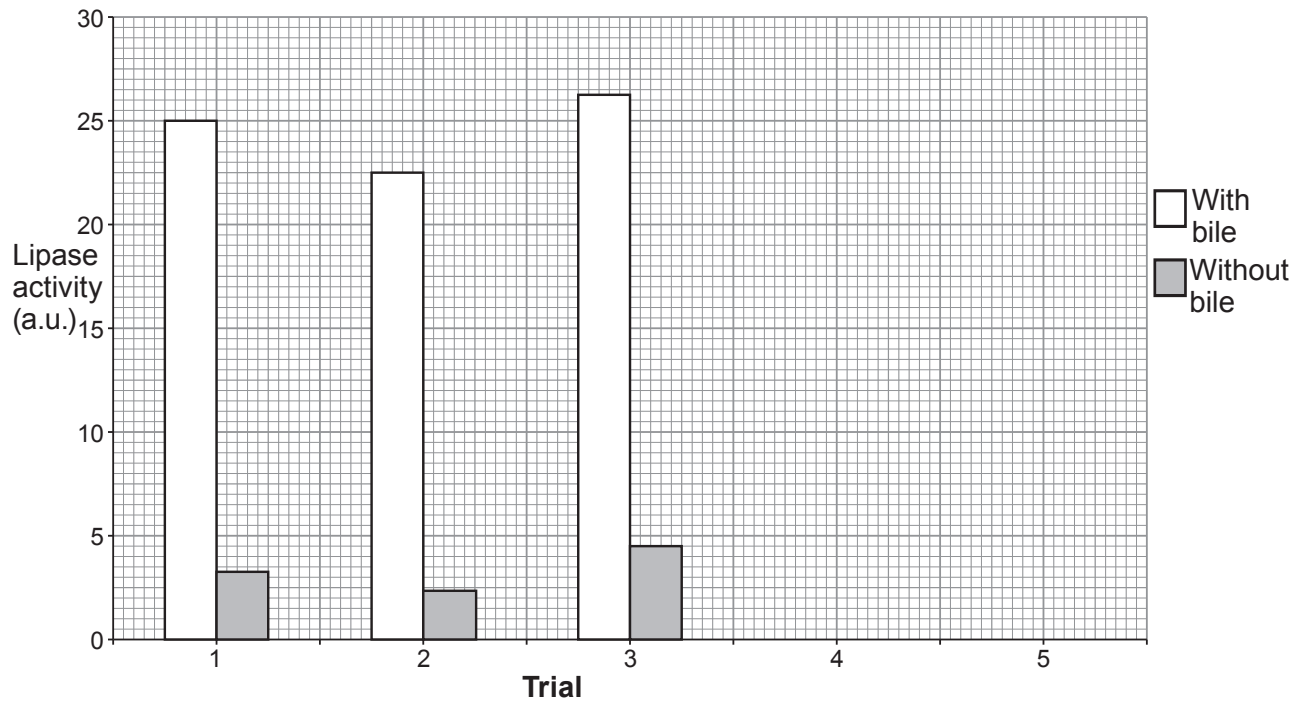
- (b) In an investigation, the effect of lipase on the breakdown of fat was measured with and without using bile. Five trials were carried out and the results are shown in **Table 4.2** and **Graph 4.3**.

Table 4.2

		Lipase activity (a.u.)	
		With bile	Without bile
Trial	1	24.9	3.1
	2	22.6	2.3
	3	26.2	4.4
	4	22.6	1.7
	5	23.7	3.5
Mean		24.0	3.0
Ratio of means = :			

- (i) I. Calculate the ratio of the means. **Write your answer in Table 4.2.** [1]
II. Complete **Graph 4.3** by **plotting the results** for trial **4** and trial **5** from **Table 4.2**. [3]

Graph 4.3



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(ii) From these results, describe the effect of bile on the activity of lipase and give **one** piece of evidence to support your answer. [2]

.....

.....

.....

.....

(iii) Each trial started at pH 7. State how the pH would change during the trial. [1]

.....

(iv) State the purpose of measuring the activity of lipase without bile in each of the trials. [1]

.....

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5. Sea anemones are animals which are often found in rock pools along the seashore. **Image 5.1** shows a sea anemone.

Image 5.1

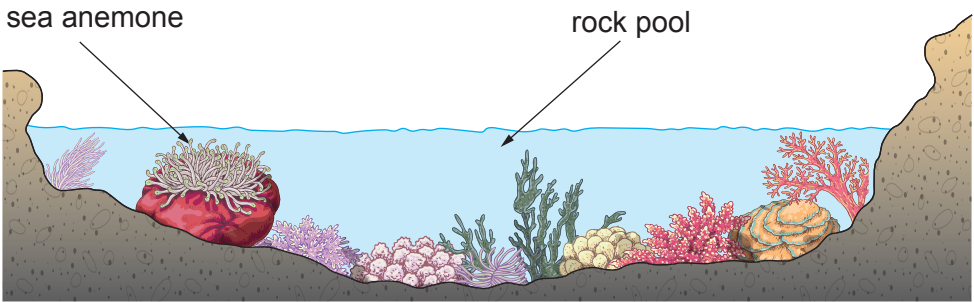


sea anemone

- On warm sunny days:
- the water temperature in rock pools increases.
 - evaporation occurs, so the concentration of salts in the water increases.
 - the water becomes acidic (pH lower than 7).
 - concentration of salt in sea water is **HIGHER** than anemone cells.
 - concentration of water in sea water is **LOWER** than anemone cells.

Image 5.2 shows a rock pool at the end of a warm day.

Image 5.2



Fact file – Conditions required for sea anemones to survive.

Oxygen concentration (mg / dm ³)	Above 9.9
pH	Between 8.1 and 9.7



Table 5.3 shows how the sea water temperature affects its oxygen concentration.

Table 5.3

Sea water temperature (°C)	Oxygen concentration (mg/dm ³)
0	14.5
10	11.2
20	9.4
30	7.6
40	4.5

Use the information to answer the following questions.

- (a) (i) State how the oxygen concentration of sea water changes when the temperature increases. [1]
-
- (ii) Calculate the mean change in oxygen concentration **per degree** when the temperature rises from 10°C to 30°C. [2]

Mean change in oxygen concentration per degree = mg/dm³

- (b) The statements in **Table 5.4** refer to conditions in a rock pool at the end of a warm, sunny day when the sea water temperature was 24°C. Write **true** or **false** for each statement. [4]

Table 5.4

Statements	True or False
Water vapour has been lost from the rock pool.	
The sea anemone has lost salt by diffusion.	
There is enough oxygen in the water for the sea anemone to survive.	
The pH of the water is suitable for the sea anemone.	
Water will pass out of the sea anemone by osmosis.	



6. (a) Enzymes are described as biological catalysts. State the function of enzymes in cells.

[1]

- (b) Gelatine is a protein which is liquid above 25°C. The gelatine protein sets when it cools down and becomes fully solid at 15°C. The gelatine protein does not set if it is broken down by an enzyme. **Image 6.1** shows a packet of gelatine.

Image 6.1

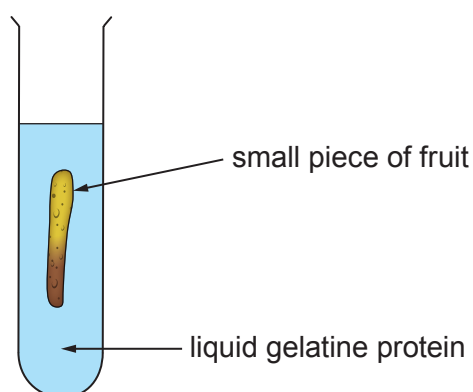


Many fruits have protease enzymes which break down proteins.

In an investigation some students used the gelatine protein to identify which fruits contain proteases.

They set up five test tubes, one of which is shown in **Image 6.2**.

Image 6.2



They placed all the tubes in a refrigerator at 5°C for some time. They then observed the tubes. Their results are shown in **Table 6.3**.

Table 6.3

Tube number	Fruit	Gelatine protein (liquid or solid)	
		At start	At end
1	fresh figs	liquid	liquid
2	fresh strawberry	liquid	solid
3	fresh kiwi fruit	liquid	liquid
4	boiled peaches	liquid	solid
5	fresh pineapple	liquid	liquid

(i) Use the information about gelatine and **Table 6.3** to answer the following questions.

I. Identify **all** the fruits which contain protease. [1]

.....

II. Explain why you reached this conclusion. [2]

.....

.....

(ii) The students' teacher commented that the result for peaches was not valid.

I. Explain the reason for this comment. [2]

.....

.....

.....

II. State how they could obtain a valid result for peaches. [1]

.....

.....

(iii) Suggest a temperature at which gelatine should be kept before pouring it into the test tubes. [1]

..... °C

(iv) State **one** variable, other than temperature, which the students should have controlled to ensure fair testing. [1]

.....

9



7. The increasing level of carbon dioxide in the air is causing global warming. **Image 7.1** shows one effect of global warming

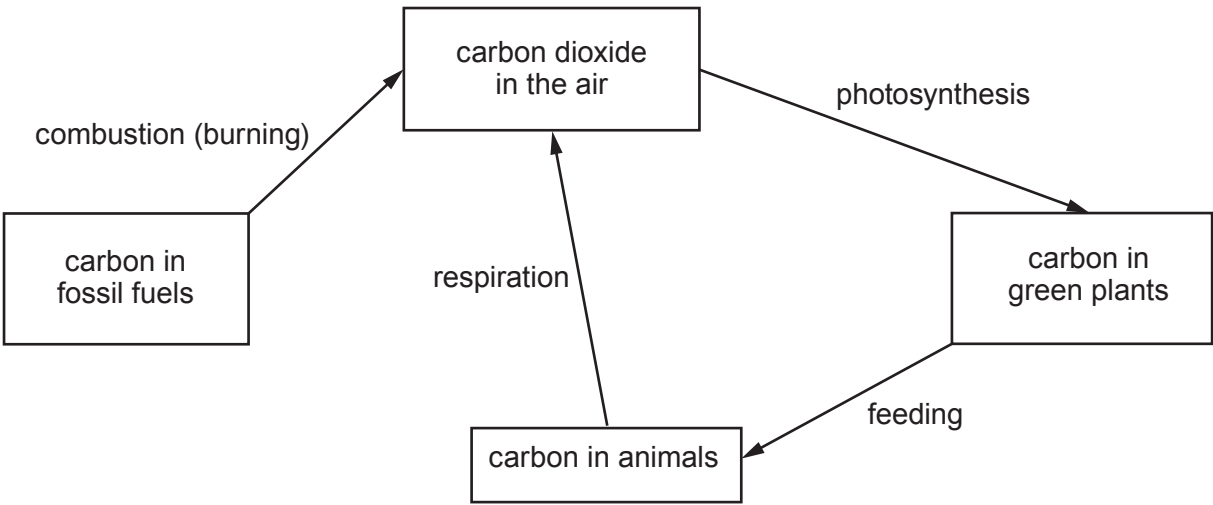
Image 7.1



Polar bear’s habitat damaged by global warming.

Image 7.2 shows part of the carbon cycle.

Image 7.2



Explain how the processes of photosynthesis, respiration and combustion each affect global warming. Suggest how humans could try to reduce the levels of carbon dioxide in the air. [6 QER]

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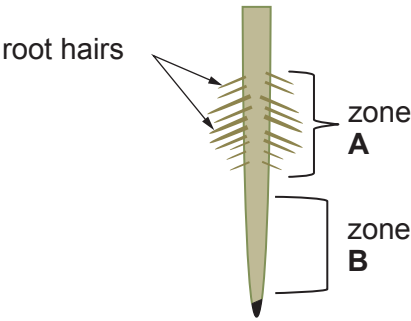
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8. Part of a plant root is shown in **Image 8.1**.

Image 8.1



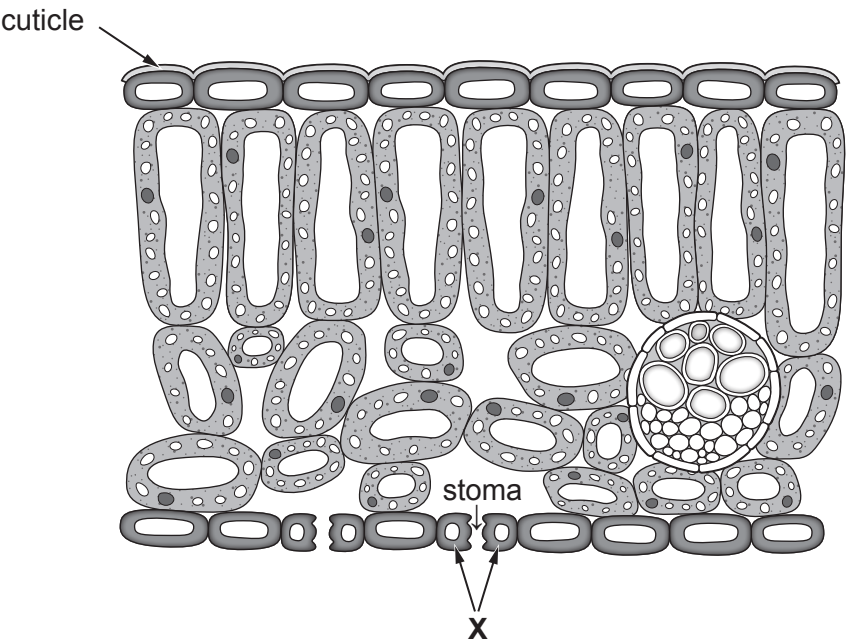
(a) Explain why water uptake in zone **A** is greater than in zone **B**. [1]

.....

.....

(b) **Image 8.2** represents a section through the leaf of a plant.

Image 8.2



(i) On **Image 8.2**, draw an arrow to show the tissue which transports water to all parts of the plant. Label the arrow with the name of the tissue. [2]



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(ii) I. Name cells **X** shown in **Image 8.2**. [1]

.....

II. State how the stoma and cuticle are involved in the control of water loss from a leaf. [2]

Stoma

.....

.....

Cuticle

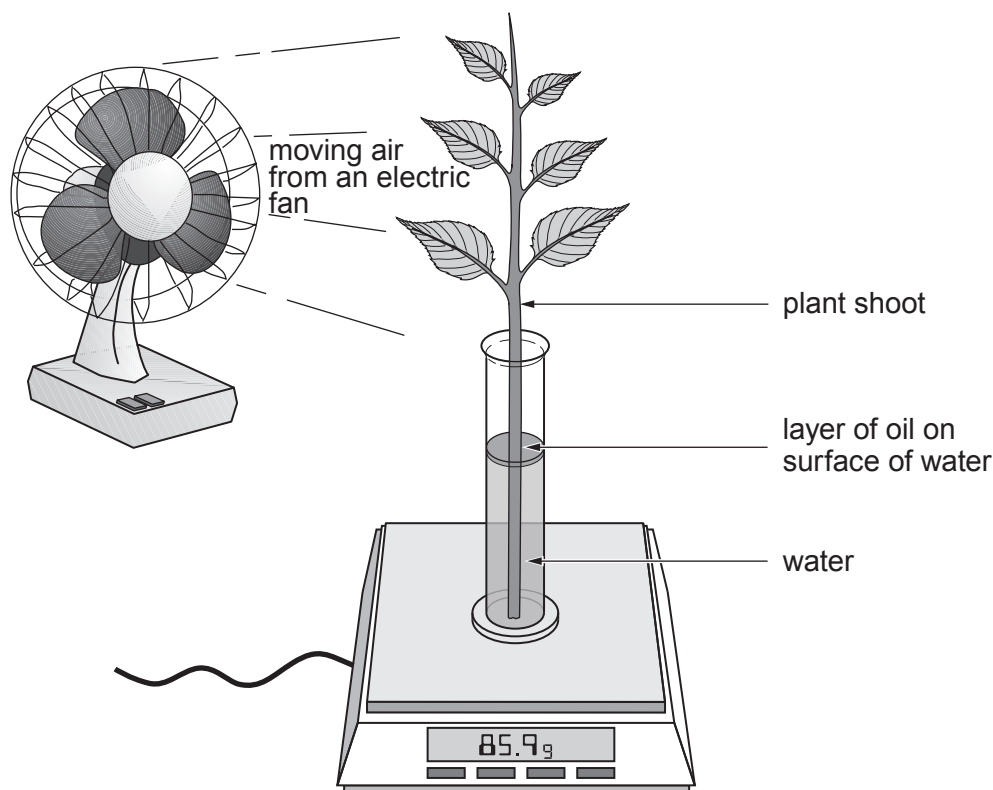
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- (c) Megan and Rhys investigated the loss of water from a leafy shoot. They used the apparatus shown in **Image 8.3**.

Image 8.3



They recorded the loss of mass after directing moving air, at different speeds, onto the shoot.

- (i) State the scientific term for the evaporation of water from the leaves of a plant. [1]

.....

- (ii) State **one** way in which Megan and Rhys could ensure that they carried out a fair test. [1]

.....

.....

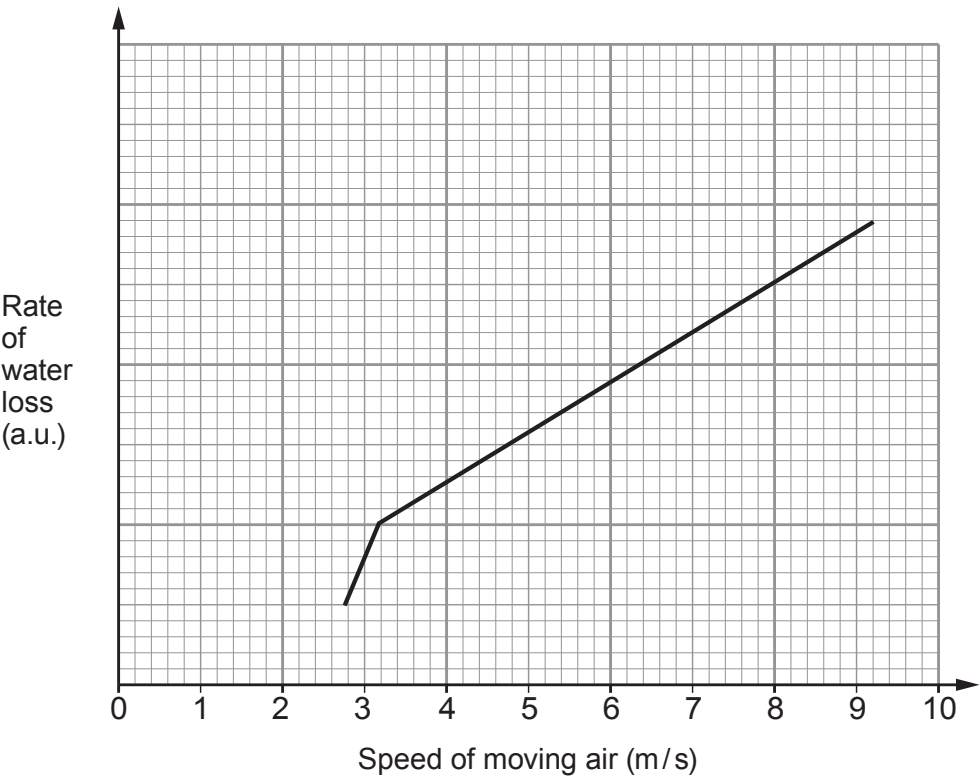
- (iii) State why it was important that the layer of oil was added. [1]

.....



(iv) The results of their investigation are summarised in **Graph 8.4**.

Graph 8.4



I. Describe the effect of increasing the speed of moving air on the rate of water loss.

[2]

.....

.....

.....

.....

II. **Sketch a line on Graph 8.4** to suggest the result you would expect if the **humidity** of the air was increased.

[1]

12



9. Image 9.1 shows chickens in two different farming systems.

Image 9.1



Free-range farm

Intensive farm

In 1950 chicken was an expensive food and most adults in the UK, on average, ate only 1100 g each year. By 2000 they ate 25 kg per year as intensive farming had made chicken much cheaper.

In intensive farming, large numbers of chickens are reared indoors. Environmental conditions and food supply are constantly controlled. The chickens grow faster than free-range chickens and use less energy as their movement is restricted.

The farmer can monitor the chickens more easily than on a free-range farm. Much less land is used and labour costs are lower but larger amounts of concentrated waste are produced.

- (a) (i) Use the information given to calculate the increase in mass in the annual consumption of chicken for a **family of four adults** in the UK between 1950 and 2000. [2]

Increase in mass = kg /family/year



During an investigation, the growth of chickens from different farm systems was compared. The results are shown in **Graph 9.2** and **Tables 9.3** and **9.4**.

Graph 9.2

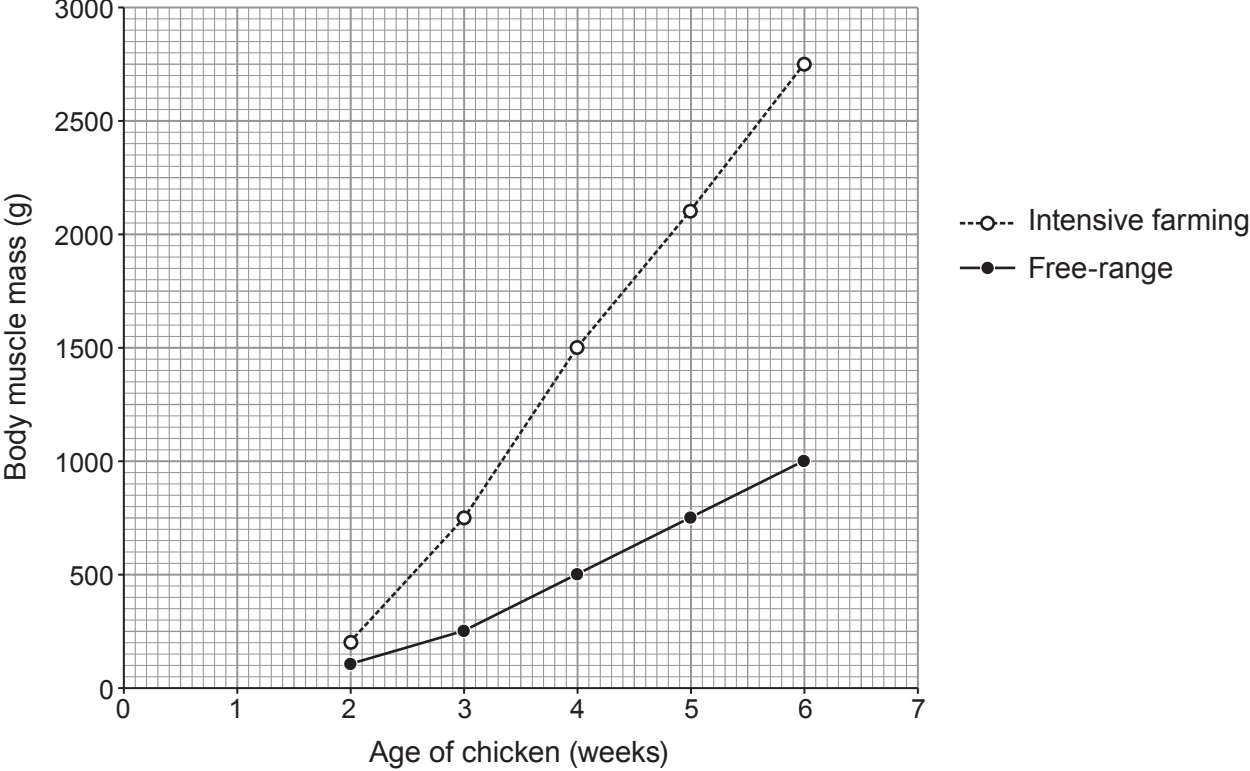


Table 9.3

Organ mass in chickens at six weeks

Organ	Mass (g)	
	Free-range	Intensively farmed
Heart	6.5	4.8
Lungs	4.0	3.5
Liver	20.0	15.0

Table 9.4

Bone quality in chickens at six weeks

	Free-range	Intensively farmed
Bone density (g/cm ³)	1.29	0.79
Presence of broken bones (%)	2.5	37
Length of leg bones (mm)	73	118



Examiner
only

- (ii) Use **Graph 9.2** to calculate the difference in the body muscle mass at 5 weeks between intensively farmed chicken and a free-range chicken. [1]

Difference = g

- (b) (i) Use the information on page 23 to suggest **one** advantage to farmers of farming chickens intensively. [1]

.....

.....

- (ii) State **one** feature of intensive farming which is an environmental disadvantage. [1]

.....

- (c) Some groups of people have ethical objections to intensive farming because of animal welfare concerns.

Using the information in **Tables 9.3** and **9.4**, state **three** features of intensive farming which support this point of view. [3]

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END OF PAPER

8

